

Ship Roll Motion Control

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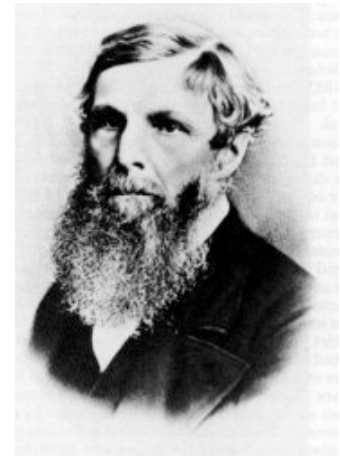
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Motivation

The effects of roll on ship performance became more noticeable in the mid-19th century when:

- Sails were replaced by engines
- Broad arranges changed to turrets

Many devices have been proposed leading to interesting control problems



W. Froude (1819-1879)

Performance can easily fall short of expectations because of deficiencies in control system design due to

- ▶ Fundamental limitations due to system dynamics
- ▶ Limited actuator authority
- ▶ Disturbances with large changes in spectral characteristics

Outline

- ▶ Ship Roll Control Devices
 - ▶ Working Principles,
 - ▶ Performance,
 - ▶ Historical aspects
- ▶ Key aspects of ship roll dynamics for control design
 - ▶ Models,
 - ▶ Wave-induced motion
 - ▶ Simulation and control design models
- ▶ Control System Design
 - ▶ Objectives,
 - ▶ Fundamental limitations,
 - ▶ Control strategies
- ▶ Research Outlook
 - ▶ Unsolved problems, new devices



I - Ship Roll Control Devices

Performance,
Working Principles,
and Historical Aspects

Ship Performance and Roll Motion

Ship roll motion

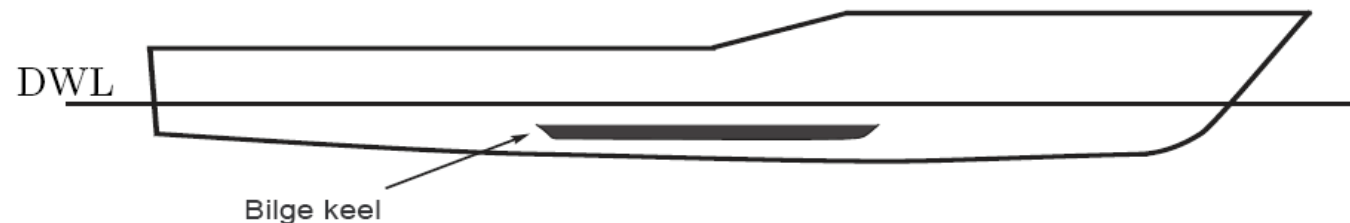
- ▶ Affects crew performance
- ▶ Can damage cargo
- ▶ May prevent the use of on-board equipment



From a ship operability point of view it is necessary to reduce not only roll angle but also roll accelerations.

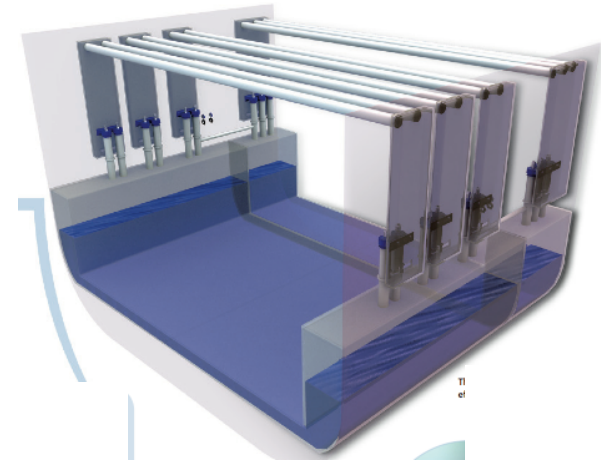
Bilge Keels

- 10to 20% roll reduction (RMS)
- Low maintenance
- No control
- No occupied space
- Low price easy to install
- Increase hull resistance when damping is not needed
- Not every vessel can be fitted with them (ice-breakers)

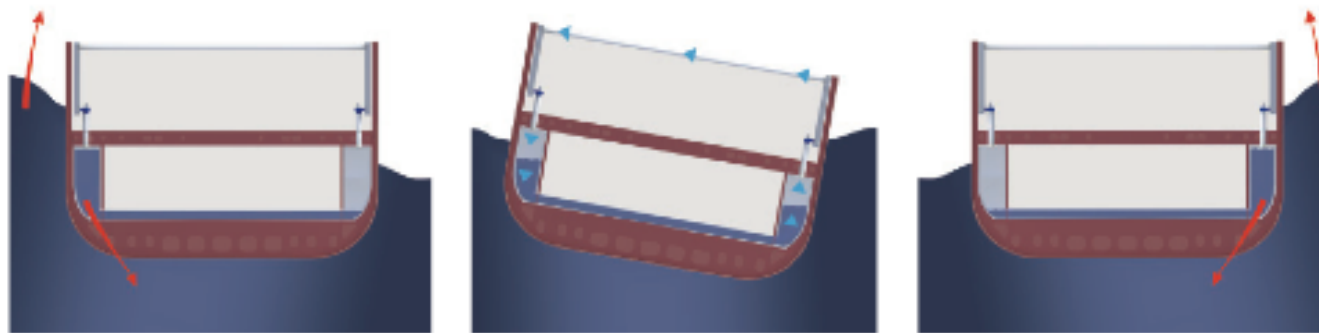


U-tanks

- 40 to 50% roll reduction (RMS)
- Active/Passive
- Independent of the vessel speed
- Anti heeling
- Heavy
- Occupy large spaces
- Affects stability due free-surface

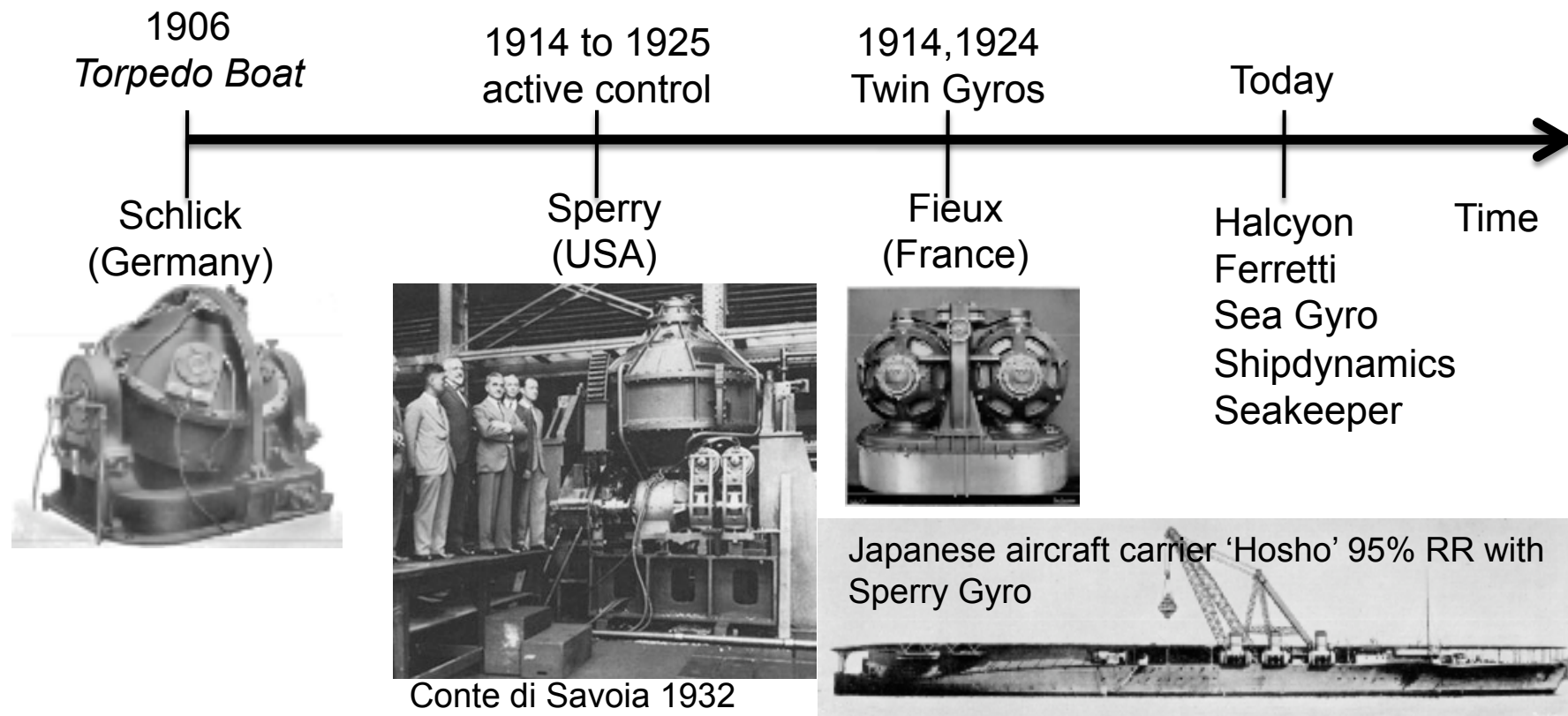


Interring Rolls-Royce



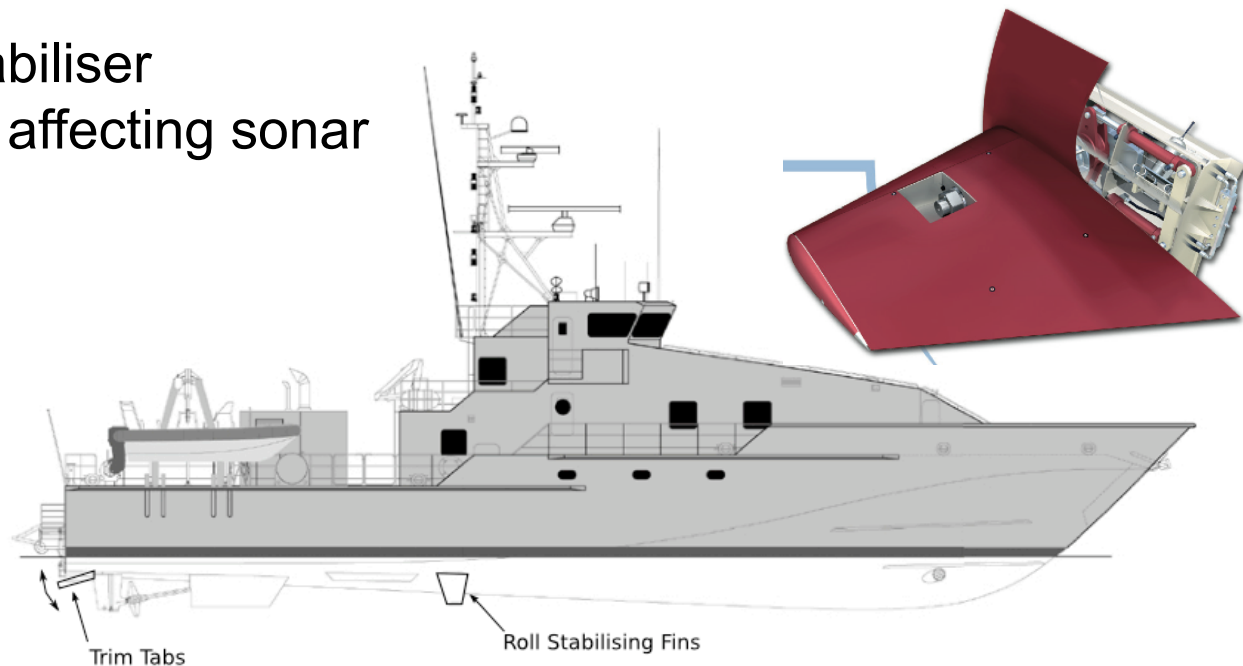
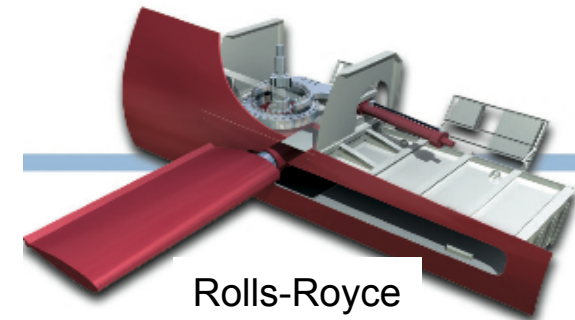
Gyro-stabilisers

- 60% to 90% roll reduction (RMS)
- Performance independent of the vessel speed



Fin-stabilisers

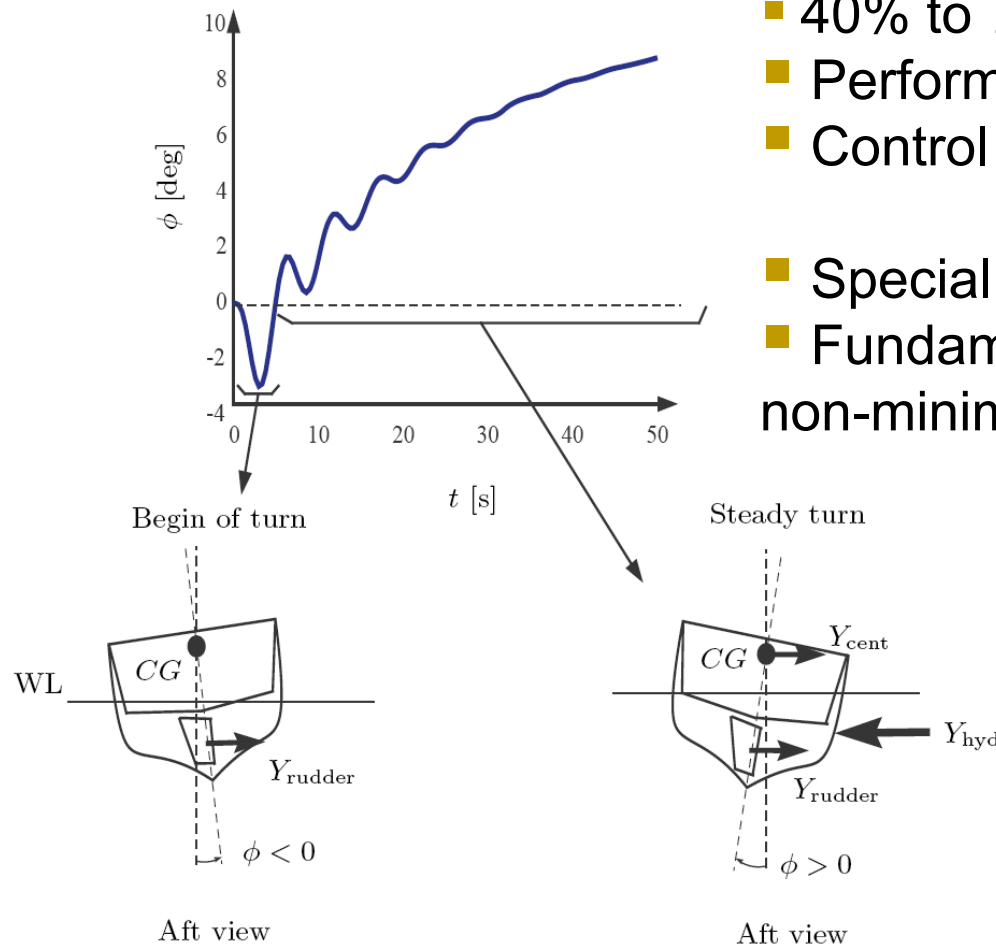
- 60% to 90% roll reduction (RMS)
- Performance depends on speed
- Control is important for performance
- Easy to damage
- Most Expensive stabiliser
- Can produce noise affecting sonar
- Dynamic stall



Rudder Roll Stabilisers (RRS)

- 40% to 70% roll reduction (RMS)
- Performance depends on speed
- Control is important for performance

- Special rudder machinery
- Fundamental limitations in control due to non-minimum phase dynamics (NMP)



Historical Aspects

Year	Device	Ship	Designer	Type
1870	Bilge keels	-	Froude (GBr)	Passive
1880	Tanks	<i>Inflexible</i>	Watt and Froude (GBr)	Passive
1891	Weight	<i>Cecile</i>	Thornycroft (GBr)	Active
1906	Gyro	<i>Sea-Bar</i>	Schlick (Ger)	Passive
1909	Weight	Steamer	Crémieu (Fra)	Passive
1910	U-tank	<i>Ypiranga</i>	Frahm (Ger)	Passive
1915	Gyro	<i>Conte di Savoia</i>	Sperry Company (USA)	Active
1924	Gyro (double wheel)	Destroyer	Fieux (Fra)	Passive
1924	Fins (variable angle)	<i>Matsu Maru</i>	Motora (Jap)	Active
1933	Fins (variable area)	<i>Aviso Estourdi</i>	Kefeli (Ita)	Active
1836	Fins (variable angle)	<i>HMS Bittern</i>	Denny-Brown (GBr)	Active
1939	U-tank	<i>Hamilton</i>	Minorsky (USA)	Active
1972	Rudder	<i>M.S. Peggy</i>	van Gunsteren (Ndl)	Active
1974	Rudder	<i>Manchester Concorde</i>	Cowley & Lambert (GBr)	Active

T. Perez (2005) *Ship Motion Control*, Springer



II - Key Aspects of Ship Roll Dynamics for Control Design

Models

Ocean Environment

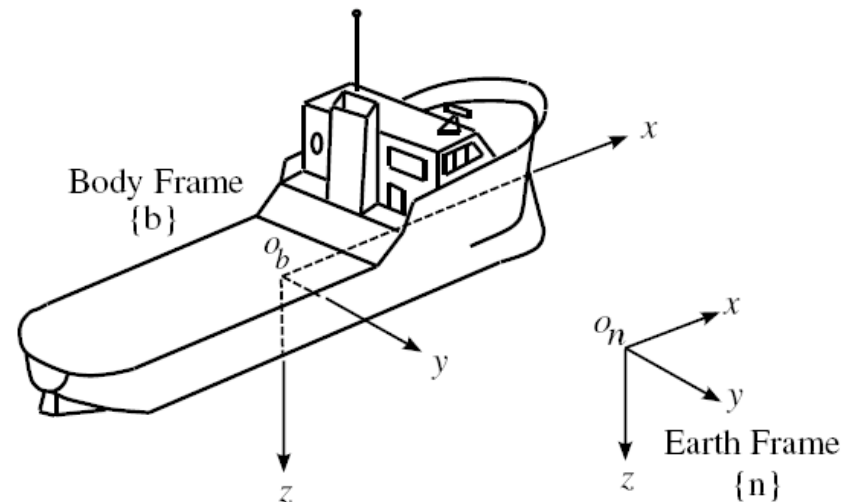
Wave-induced motion

Dynamics of Roll Motion

General model form to describe ship motion:

$$\boldsymbol{\eta} \triangleq \begin{bmatrix} \mathbf{p}_{b/n}^n \\ \boldsymbol{\Theta} \end{bmatrix} = [N, E, D, \phi, \theta, \psi]^T.$$

$$\boldsymbol{\nu} \triangleq \begin{bmatrix} {}^n \dot{\mathbf{p}}_{b/n}^b \\ \boldsymbol{\omega}_{b/n}^b \end{bmatrix} = [u, v, w, p, q, r]^T.$$

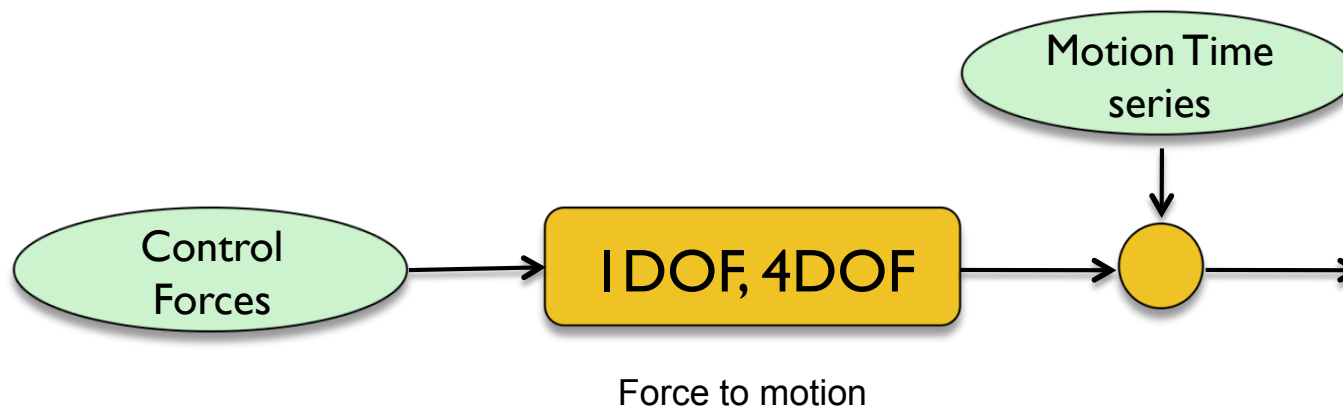


Kinematic model: $\dot{\boldsymbol{\eta}} = \mathbf{J}(\boldsymbol{\eta})\boldsymbol{\nu}$

Kinetic model: $\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau}$

Models for Control Design

For control design, output disturbance models are usually adopted:



For roll motion control problems this model is simplified to

- 1DOF: roll
- 4DOF: surge, roll, sway, and yaw

Dynamics of Roll Motion—1DOF Model

Consider only roll motion,

Kinematic model:

$$\dot{\phi} = p,$$

Kinetic model:

$$I_{xx} \dot{p} = K_h + K_c + K_d$$

Hydrodynamic
moments

Control
moments

Disturbance
moments

Hydrodynamic Moments:

$$K_h \approx \underbrace{K_{\dot{p}} \dot{p} + K_p p}_{\text{Potential Effects}} + \underbrace{K_{p|p|} p|p|}_{\text{Viscous Effects}} + \underbrace{K(\phi)}_{\text{Hydrostatic Effects}}$$

Potential
Effects

Viscous
Effects

Hydrostatic
Effects

Dynamics of Roll Motion—4DOF

Motion Variables:

$$\boldsymbol{\eta} = [\phi \ \psi]^T,$$
$$\boldsymbol{\nu} = [u \ v \ p \ r]^T,$$
$$\boldsymbol{\tau}_i = [X_i \ Y_i \ K_i \ N_i]^T,$$

Kinematic model:

$$\dot{\phi} = p, \quad \dot{\psi} = r \cos \phi \approx r$$

Kinetic model:

$$\mathbf{M}_{RB} \dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_h + \boldsymbol{\tau}_c + \boldsymbol{\tau}_d$$

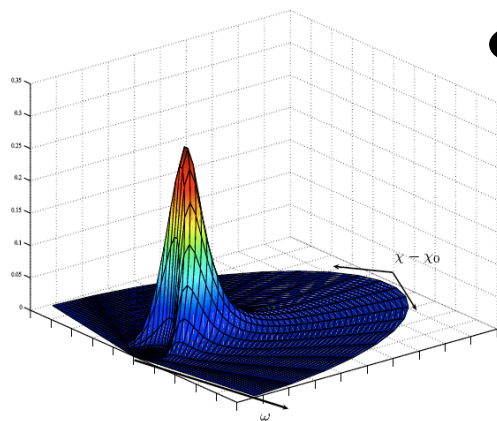
$$\boldsymbol{\tau}_h \approx -\mathbf{M}_A \dot{\boldsymbol{\nu}} - \mathbf{C}_A(\boldsymbol{\nu})\boldsymbol{\nu} - \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} - \mathbf{K}(\phi)$$

Ocean Environment

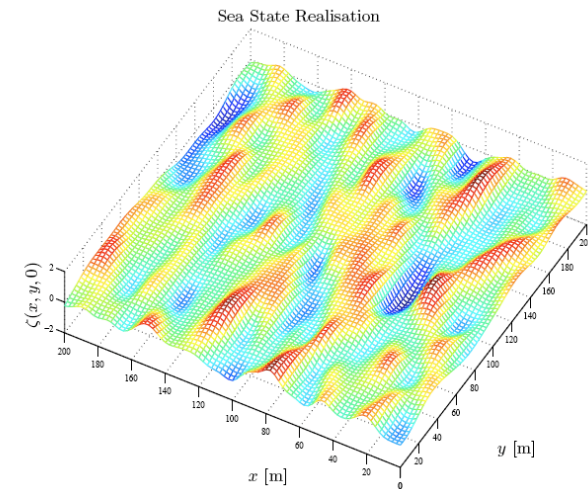
Usual assumptions for sea surface elevation $\zeta(t)$:

- ▶ *Zero-mean*
- ▶ *Gaussian (depth dependent)*
- ▶ *Narrow banded*
- ▶ *Stationary (20min to 3 hours)*

All necessary information is then in the wave elevation power spectral density (Sea Spectrum):



$$\Phi_{\zeta\zeta}(\omega, \chi)$$

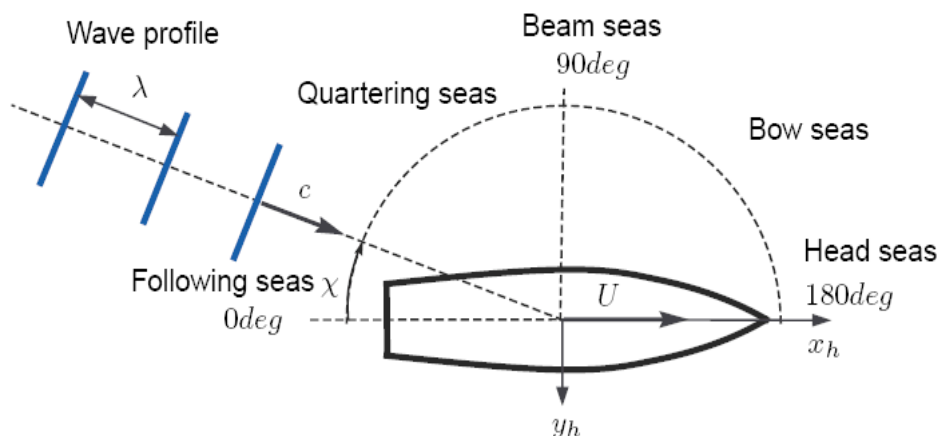
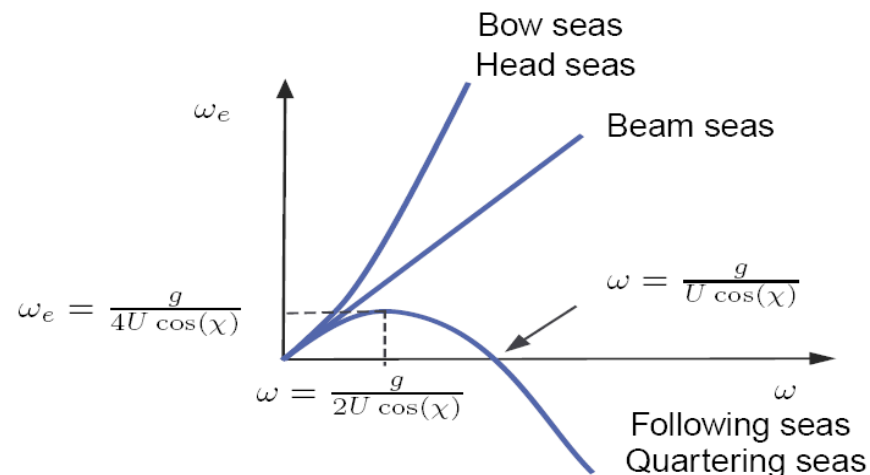


Sailing Condition and Encounter Spectrum

Sailing Condition

- ▶ *Speed*
- ▶ *Encounter angle*

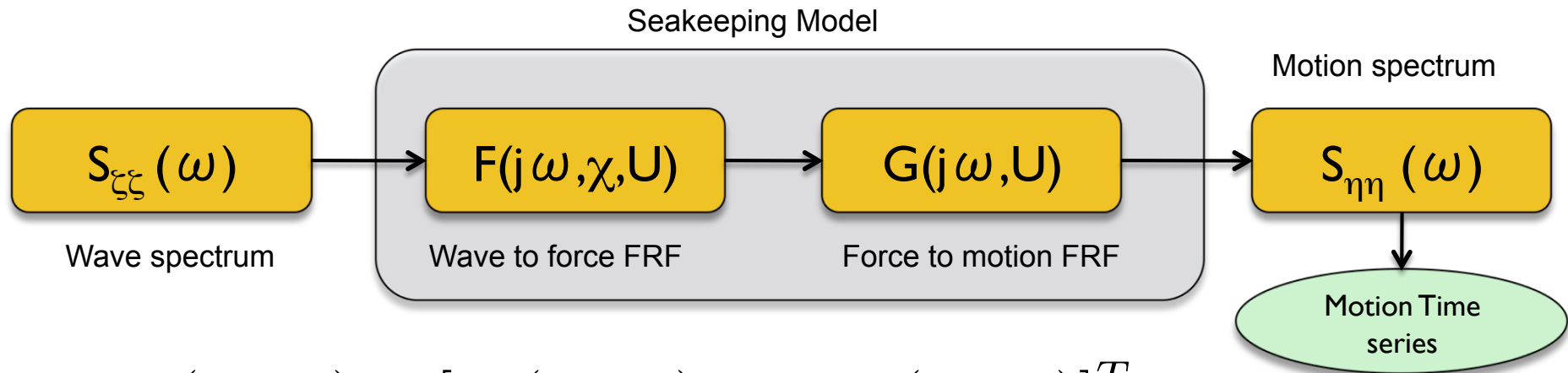
Encounter Frequency



$$\omega_e = \omega - \frac{\omega^2 U}{g} \cos(\chi)$$

$$\Phi_{\zeta\zeta}(\omega_e) = \frac{\Phi_{\zeta\zeta}(\omega)}{\left| 1 - \frac{2\omega U}{g} \cos(\chi) \right|}$$

Wave-induced Motion Motion

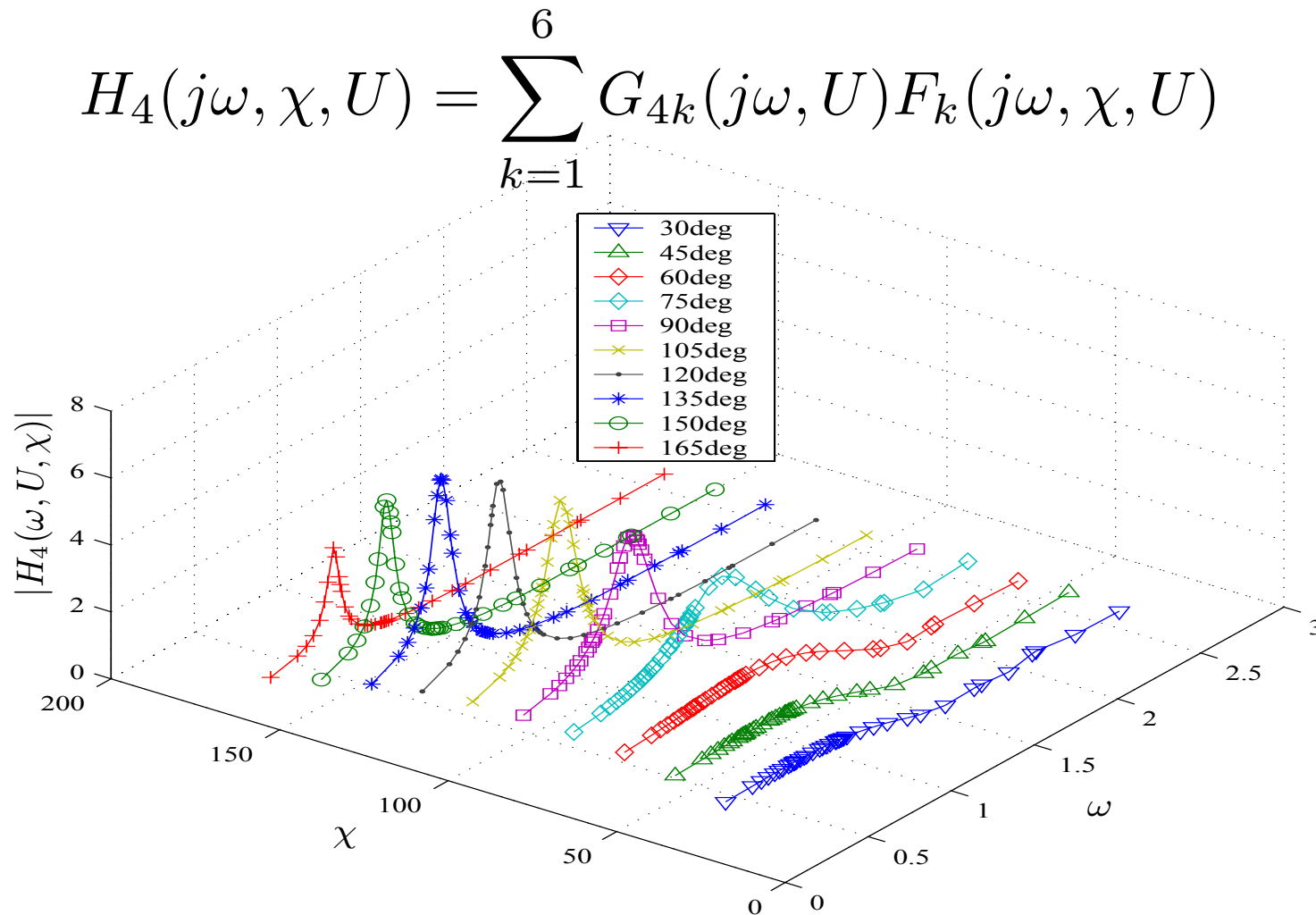


$$\mathbf{F}(j\omega, \chi) = [F_1(j\omega, \chi), \dots, F_6(j\omega, \chi)]^T$$

$$\mathbf{G}(j\omega, U) = \begin{bmatrix} G_{11}(j\omega, U) & \cdots & G_{16}(j\omega, U) \\ \vdots & & \vdots \\ G_{61}(j\omega, U) & \cdots & G_{66}(j\omega, U) \end{bmatrix} = (-[\mathbf{M}_{RB} + \mathbf{A}(\omega)]\omega^2 + j\omega\mathbf{B}(\omega) + \mathbf{G})^{-1}$$

$$\text{Roll RAO: } H_4(j\omega, \chi, U) = \sum_{k=1}^6 G_{4k}(j\omega, U) F_k(j\omega, \chi, U)$$

Example Roll RAOs Naval Vessel @15kt



III – Motion Control Design



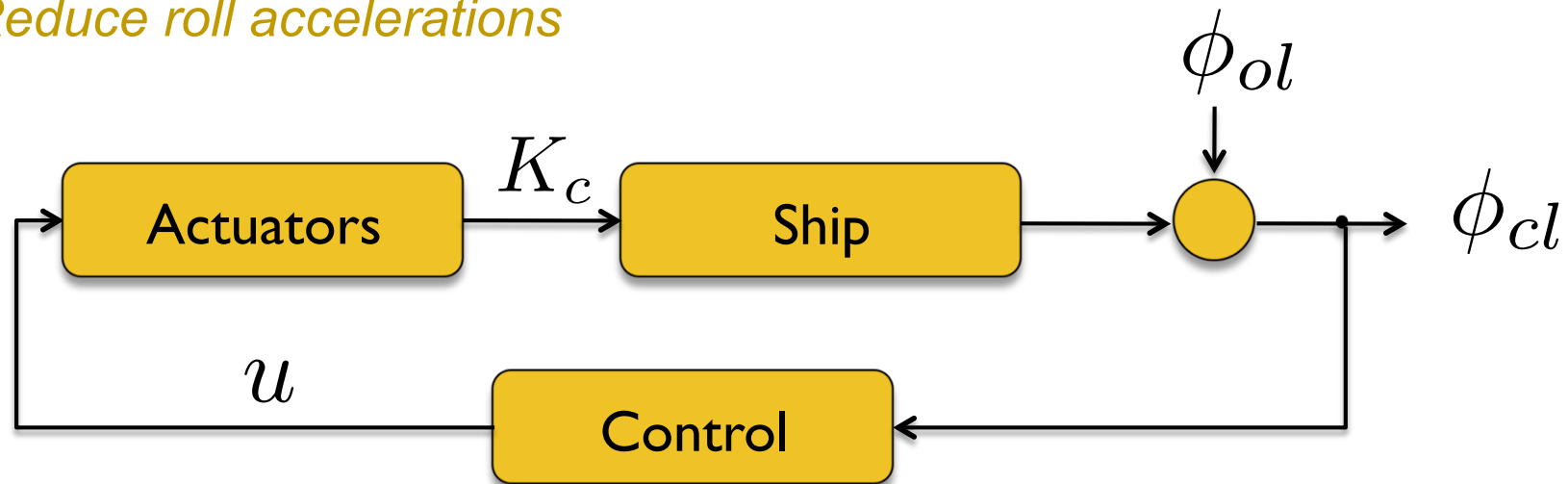
Objectives,
Performance Limitations,
and Control Strategies

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Control Design and Performance Limitations

From the ship performance point of view, the control objectives are

- ▶ *Reduce roll angle*
- ▶ *Reduce roll accelerations*



Output sensitivity

$$S(s) \triangleq \frac{\phi_{cl}(s)}{\phi_{ol}(s)}$$

Roll spectrum relations

$$\Phi_{cl}(\omega) = |S(j\omega)|^2 \Phi_{ol}(\omega)$$

Control Design and Performance Limitations

If the closed-loop system is stable minimum phase and strictly proper, the following Bode integral constraint applies

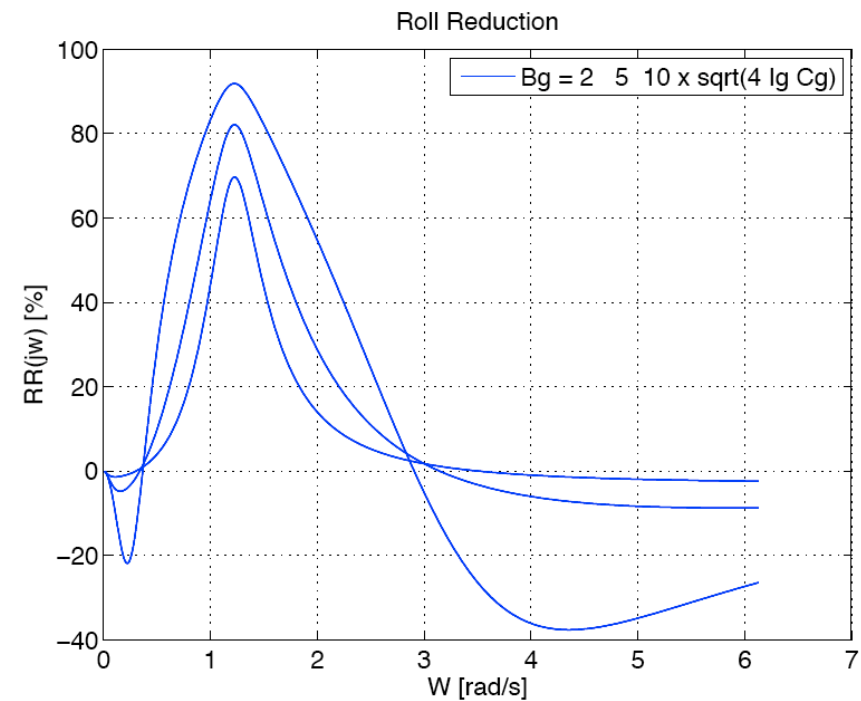
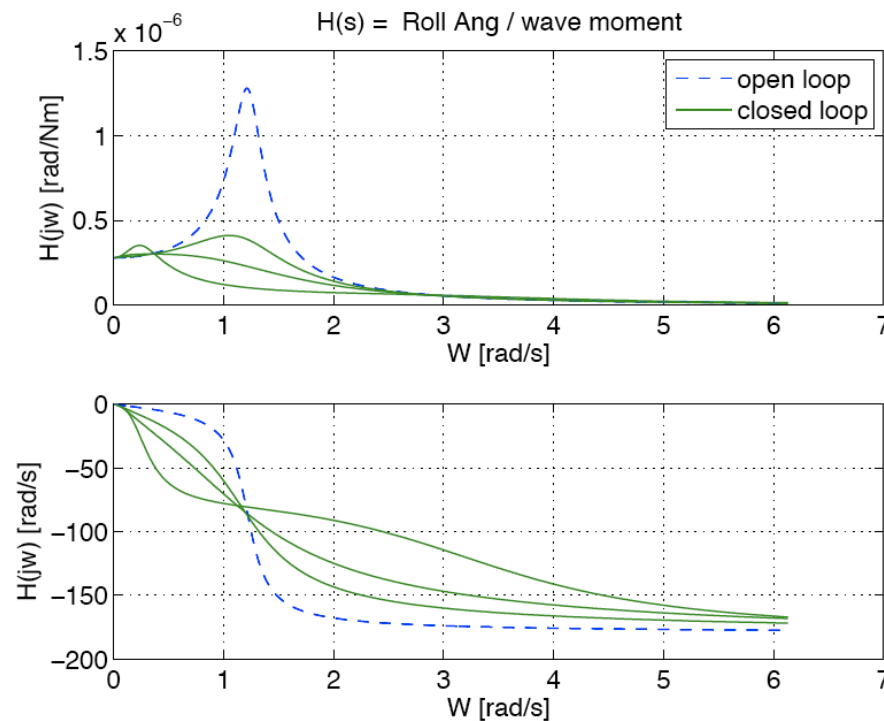
$$\int_0^{\infty} \log |S(j\omega)| d\omega = 0$$

$$\Phi_{cl}(\omega) = |S(j\omega)|^2 \Phi_{ol}(\omega)$$

$$RR(\omega) = 1 - |S(j\omega)| = \frac{|\phi_{ol}(j\omega)| - |\phi_{cl}(j\omega)|}{|\phi_{ol}(j\omega)|}$$

Example Performance Limitations

Gyrostabiliser (Perez & Steinmann, 2009)



Non-minimum phase Dynamics

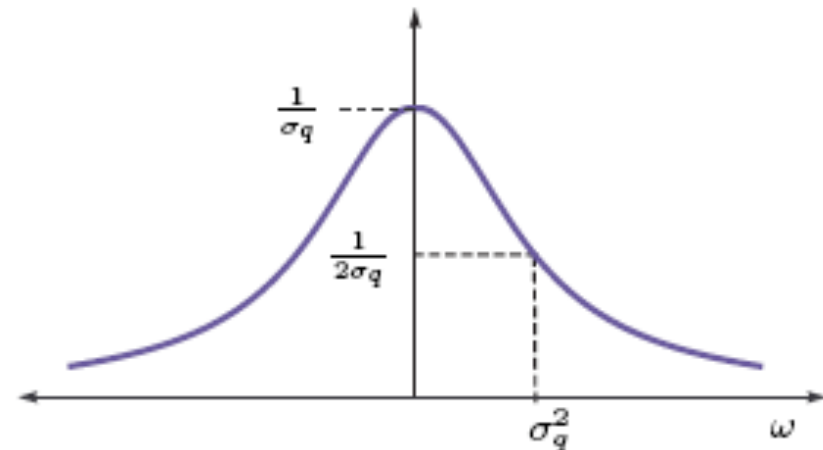
In some cases, the location of the actuators may result in NMP dynamics (with a real zero at $s=q$.)

Then, we have a Poisson-integral constrain:

$$\int_{-\infty}^{\infty} \log |S(j\omega)| W(q, \omega) d\omega = 0$$

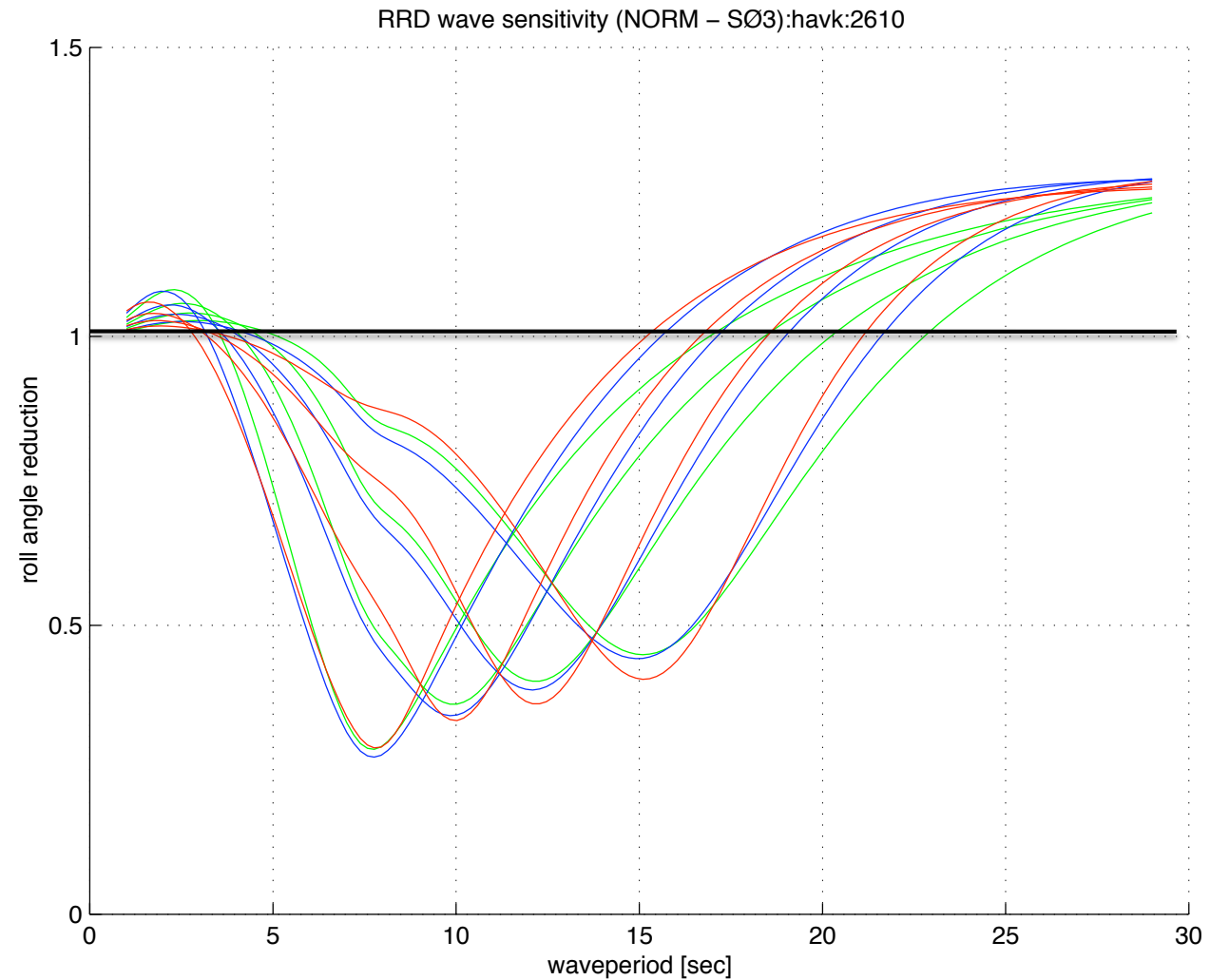
$$q = \sigma_q + j0$$

$$W(q, \omega) = \frac{\sigma_q}{\sigma_q^2 + \omega^2}$$



Non-minimum phase Dynamics

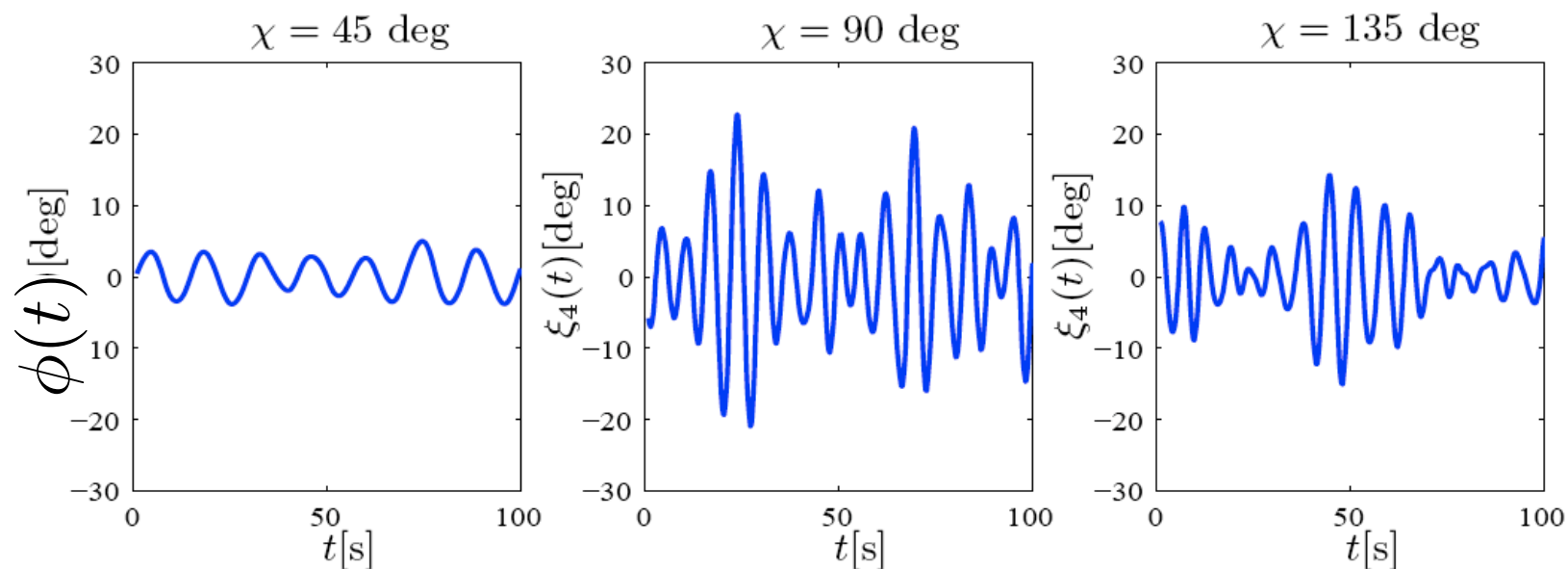
Example Blanke
et al (2000)



Energy of Wave Excitation

The energy of the wave excitation changes with the sea-state and sailing conditions (speed and heading)

For example a change in encounter angle can shift the energy significantly:



Fin Stabilisers (minimum phase case)

IDOF model including fin forces:

$$\dot{\phi} = p,$$

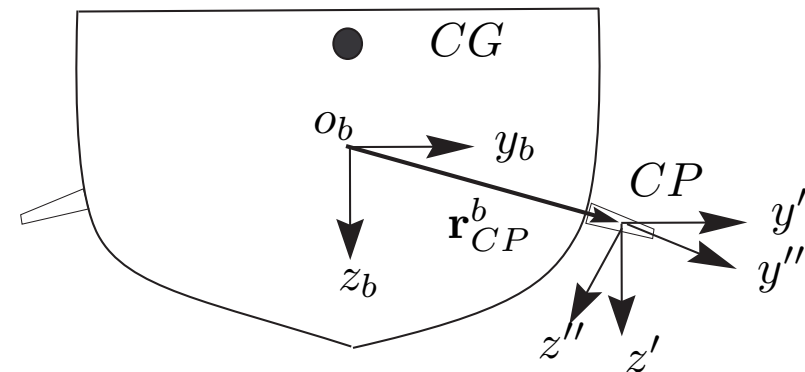
$$[I_{xx} + K_{\dot{p}}] \dot{p} + (K_p + 2 r_f K_{\alpha} U) p + K_{\phi} \phi = K_w - 2U^2 K_{\alpha} \alpha$$

Control issues:

- Parametric uncertainty
- Sensitivity-integral constraints

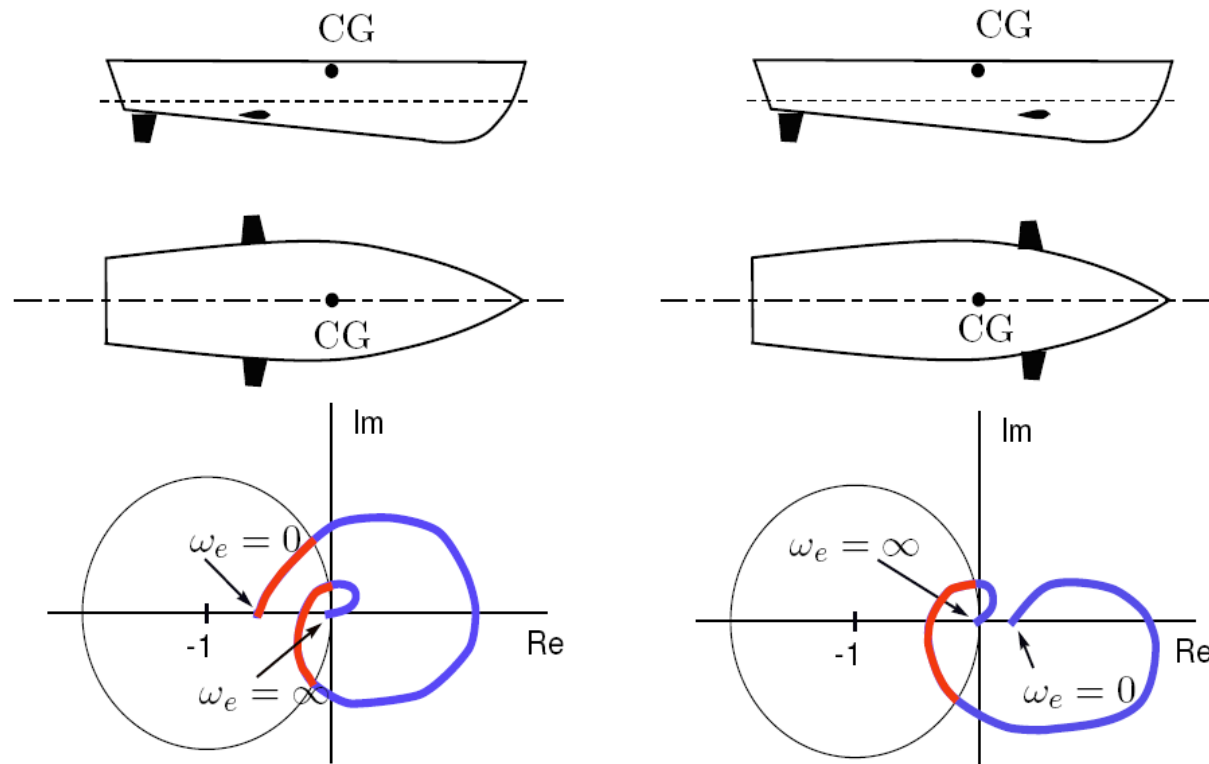
Control strategies:

- PID, Hinf



Fin Stabilisers and NMP Dynamics

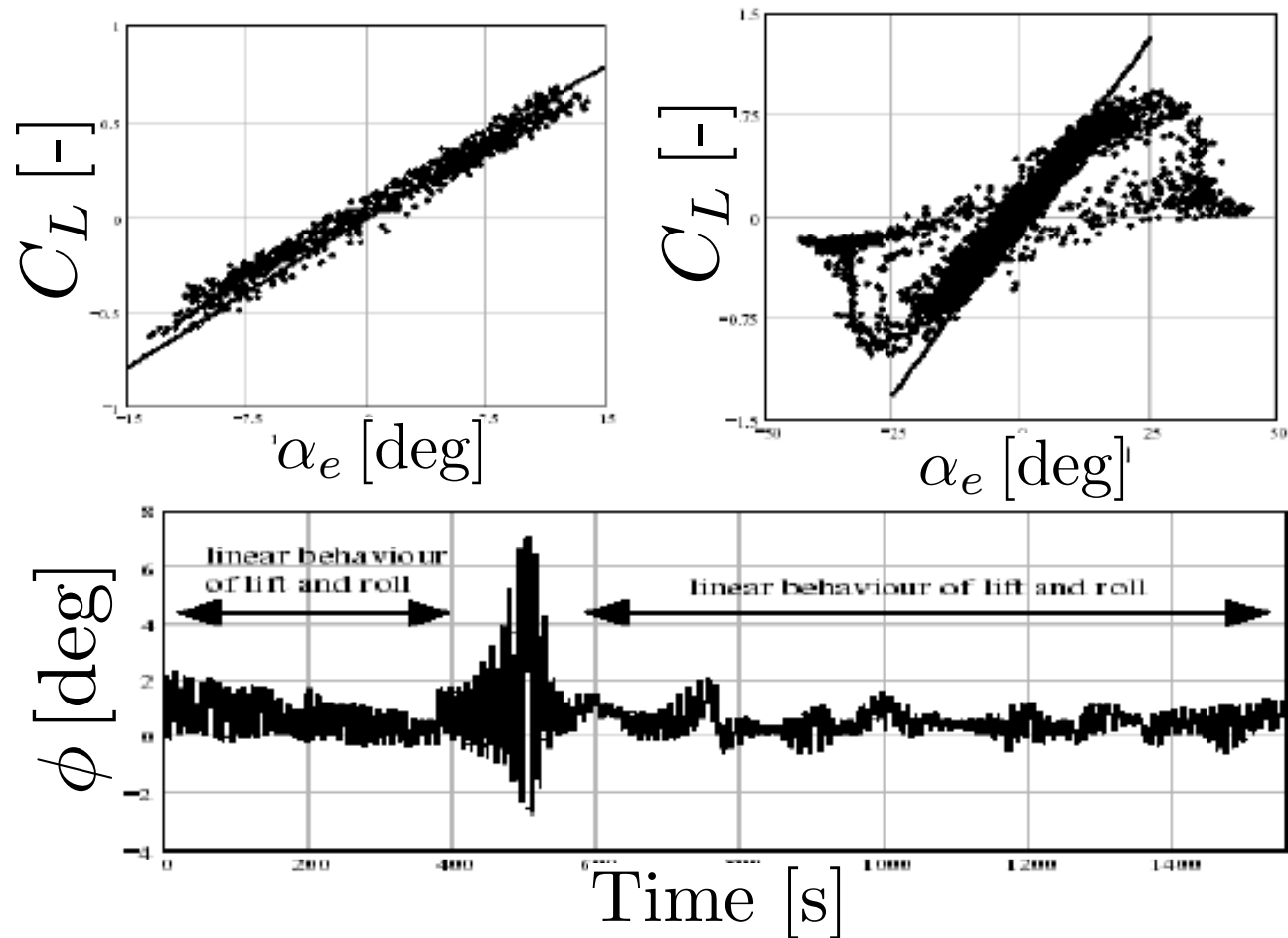
If the response from the fins is NMP, then the IDOF cannot be used, in this case roll-sway-yaw interactions need to be considered to avoid large roll amplifications at low frequencies.



Observations about fin NMP dynamics were made by Lloyd (1989).

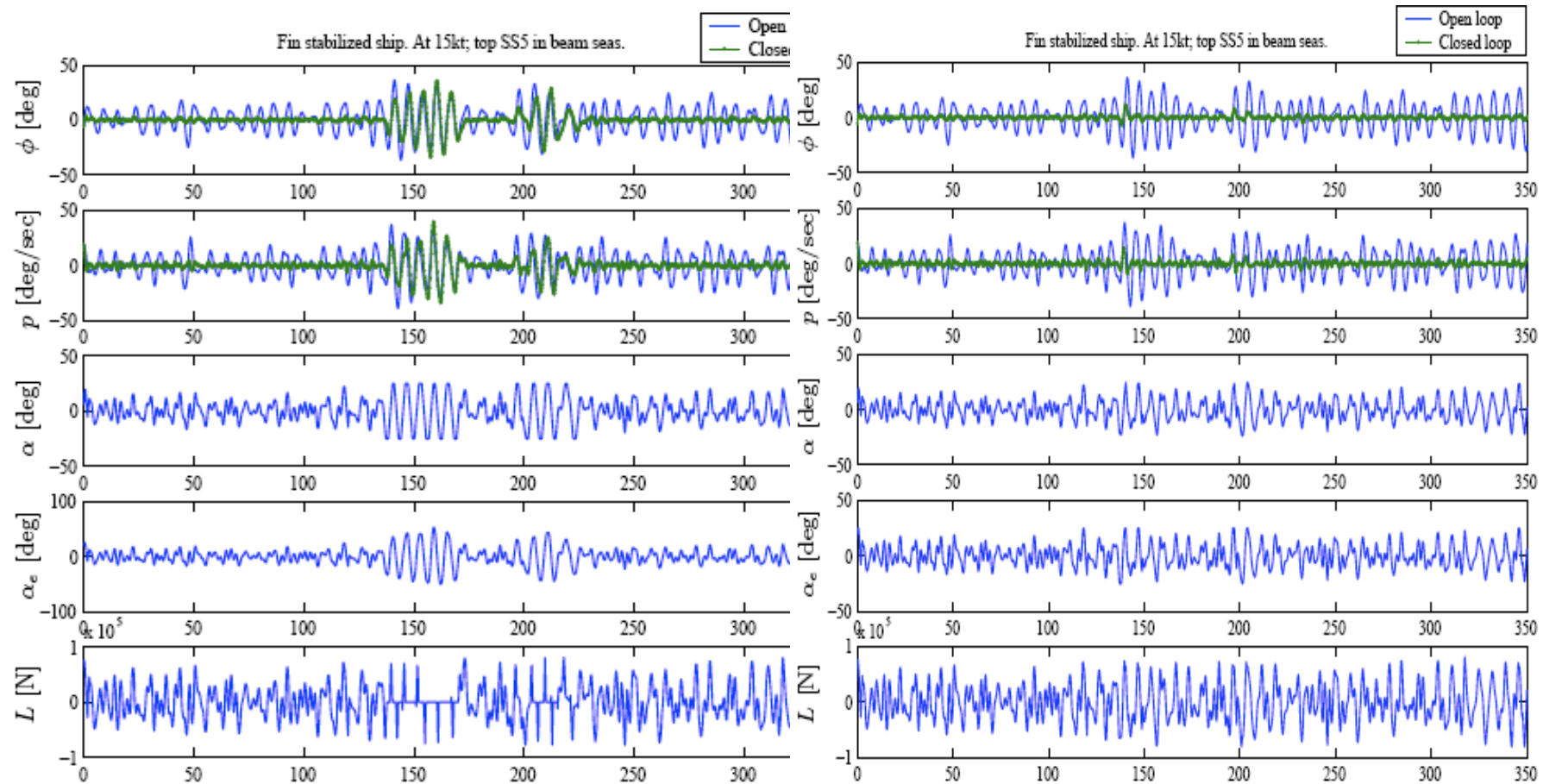
Fin Stabilisers – Dynamic Stall

Experimental results of Galliarde (2002) (MARIN, Ned)



Fin Stabilisers

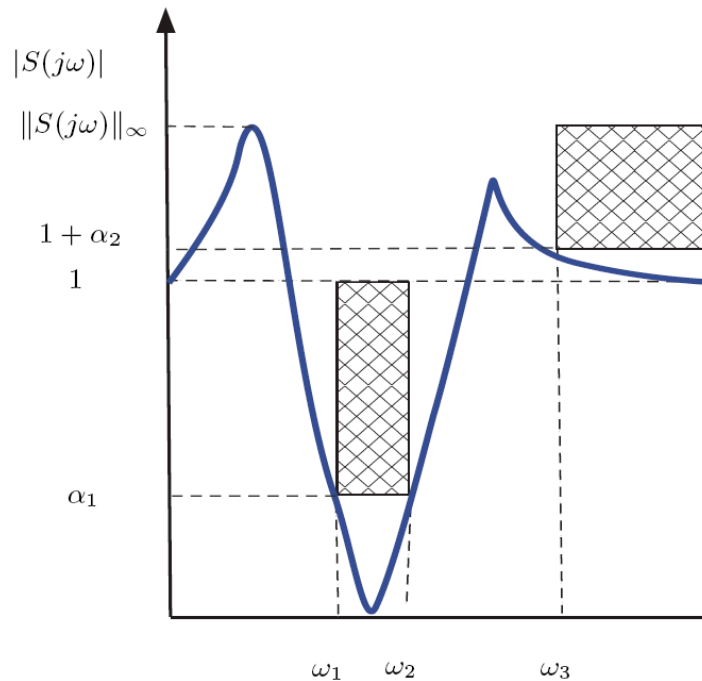
Perez & Goodwin (2003): MPC constraints effective angle of attack.



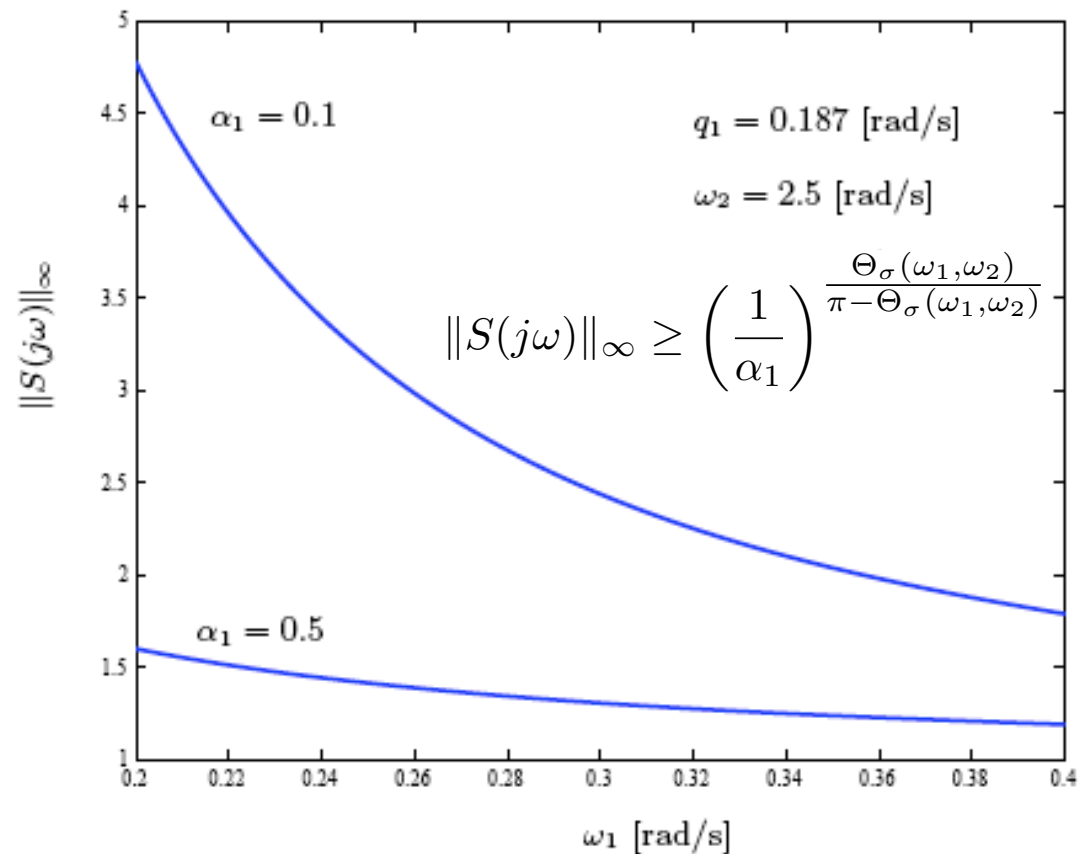
Rudder Roll Damping

- ▶ Potential discovered from autopilot without wave filter (Taggart 1970)
- ▶ Results reported by van Gunsteren in 1972 (the Netherlands)
- ▶ Cowley & Lambert (1972) used roll fbk, reported yaw interference.
- ▶ Carley & Duberley (1972) integrated rudder-fin control
- ▶ Carley (1975) & Lloyd (1975) recognise limitations of NMP dynamics
- ▶ Baitis et al.(1983) highlighted need of adaptation
- ▶ Advent of Computers in 1980 resulted in several successful results
 - ▶ Netherlands, Denmark, Sweden, United Kingdom
- ▶ Hearn & Blanke (1998) limitations using the Poisson Integral
- ▶ Perez (2003) analysed min variance and RR vs yaw interference

Performance Limitations (Hearns & Blanke, 1998)



If the controller does not adapt, we can easily have roll amplification

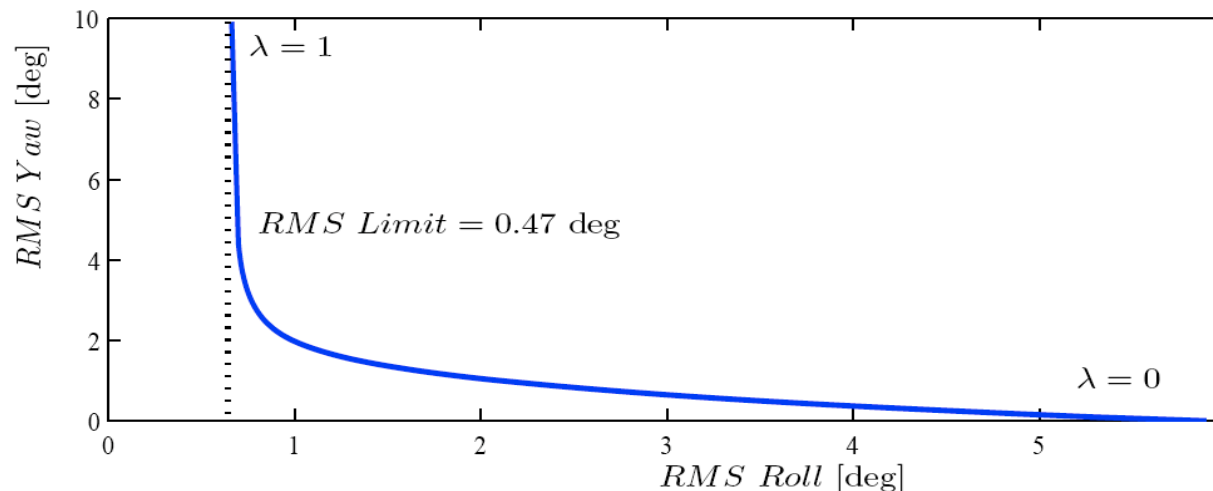


Performance Limitations – Perez et al. (2003)

Limiting Optimal Control with full knowledge of wave spectrum:

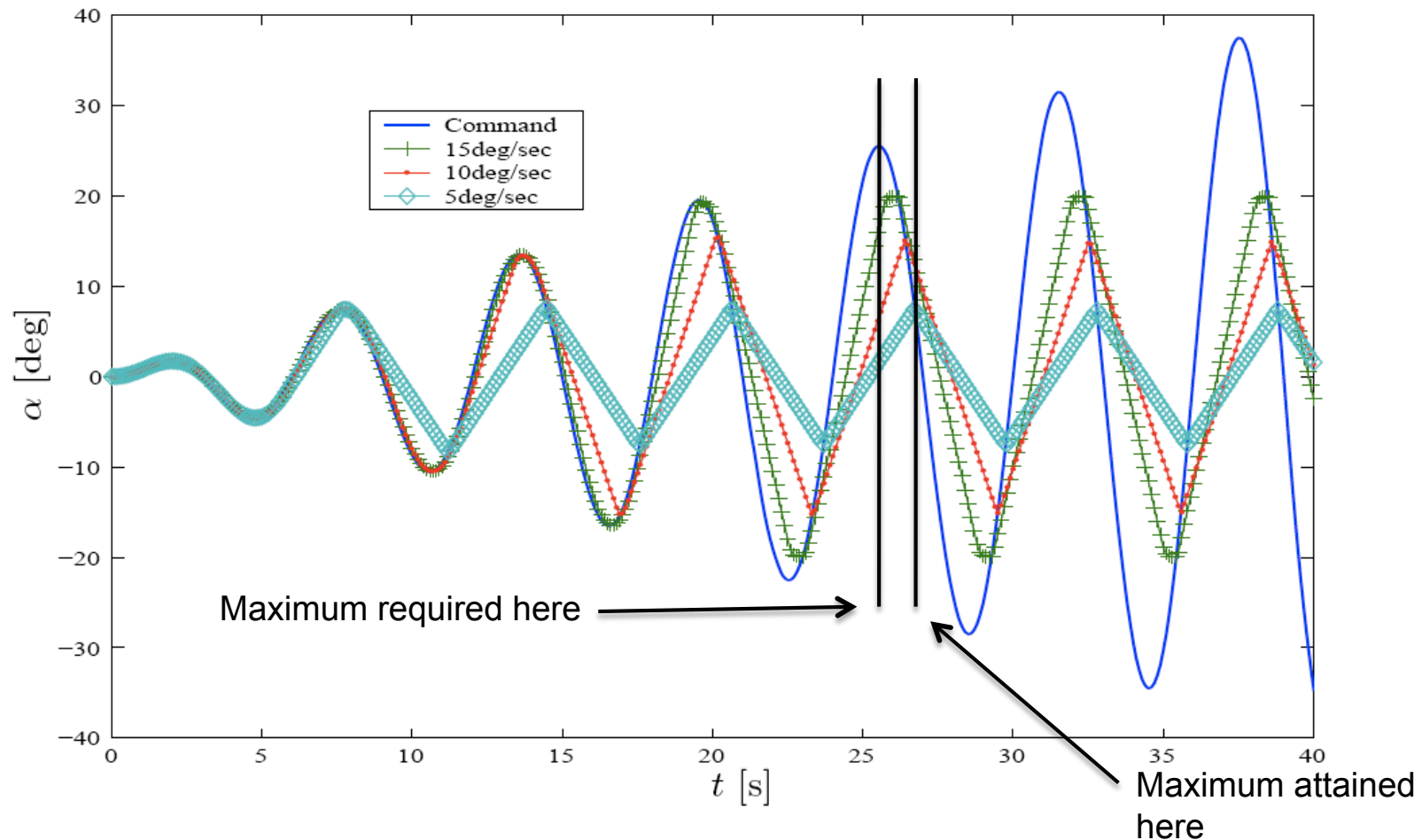
$$J = E[\lambda\phi^2 + (1 - \lambda)(\psi - \psi_d)^2]$$

Minimum variance case ($\lambda=1$): $E[\phi^2] \geq 2q\Phi_{\phi\phi}(q)$

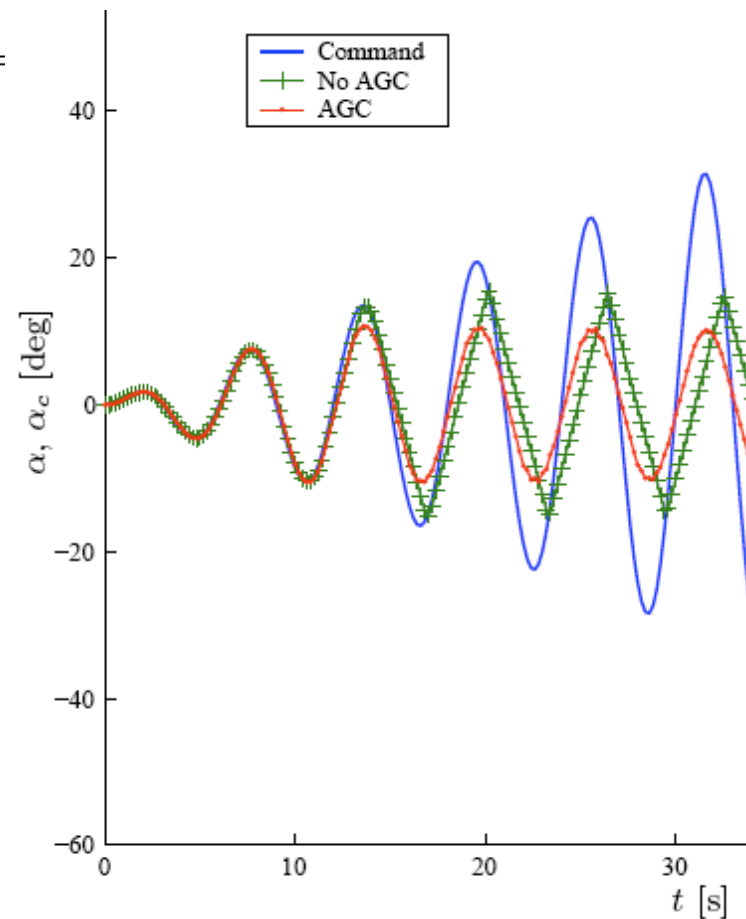
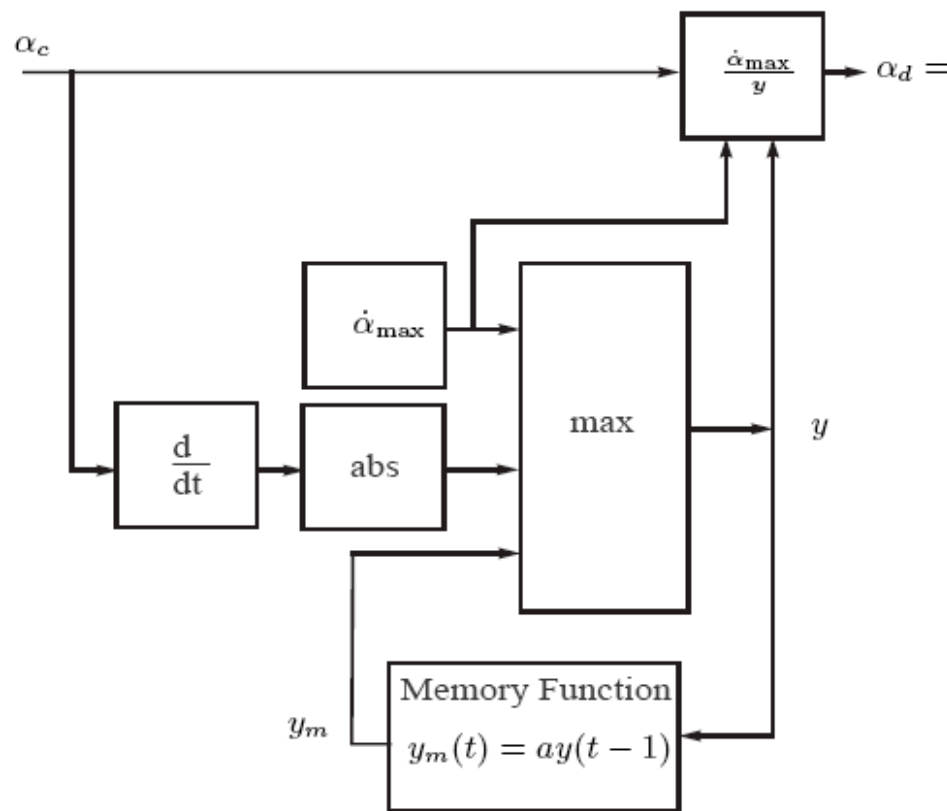


Actuator limitations on RRD

- ▶ Actuator rate limits may affect performance

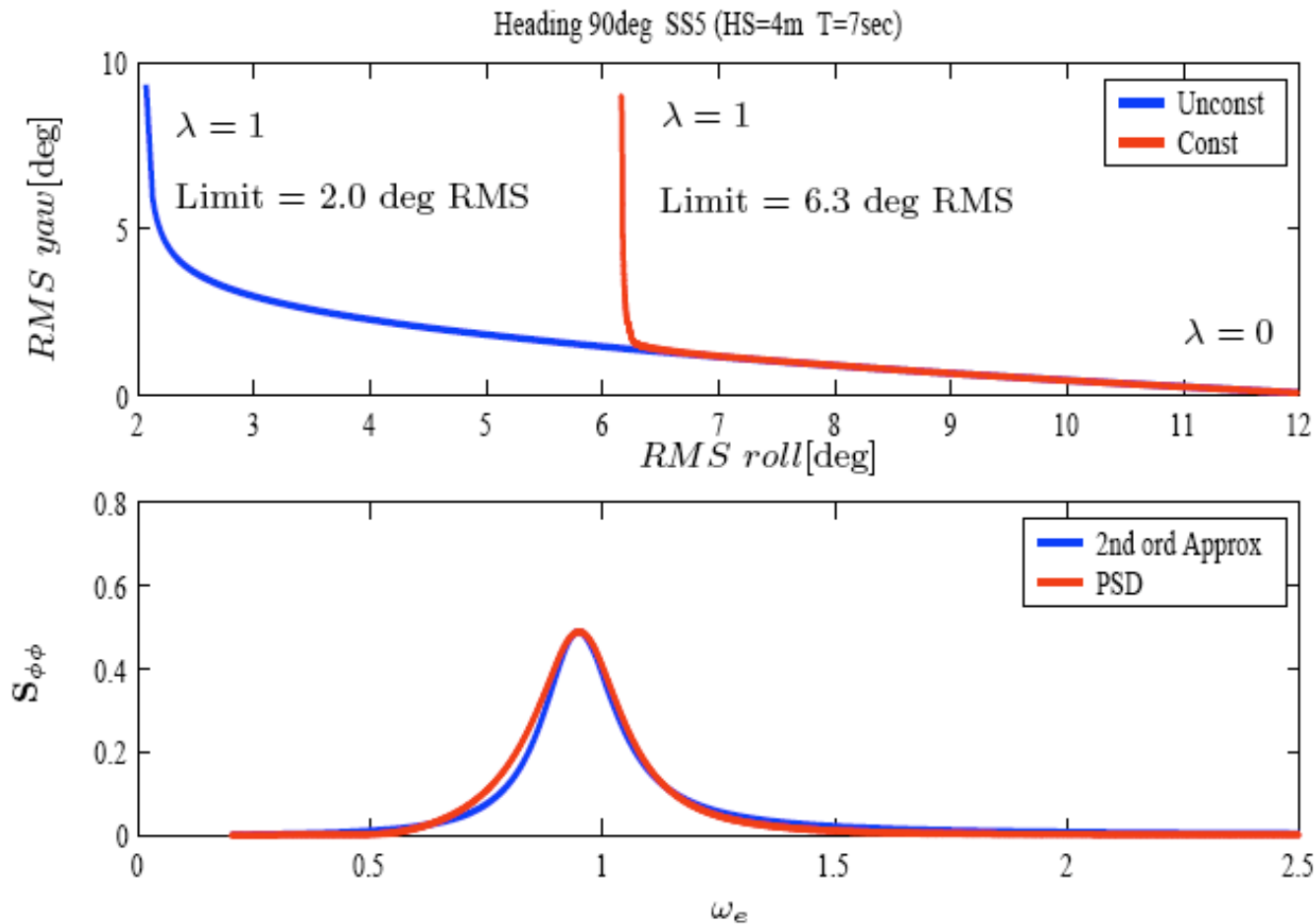


AGC – Van Amerongen et al (1982)



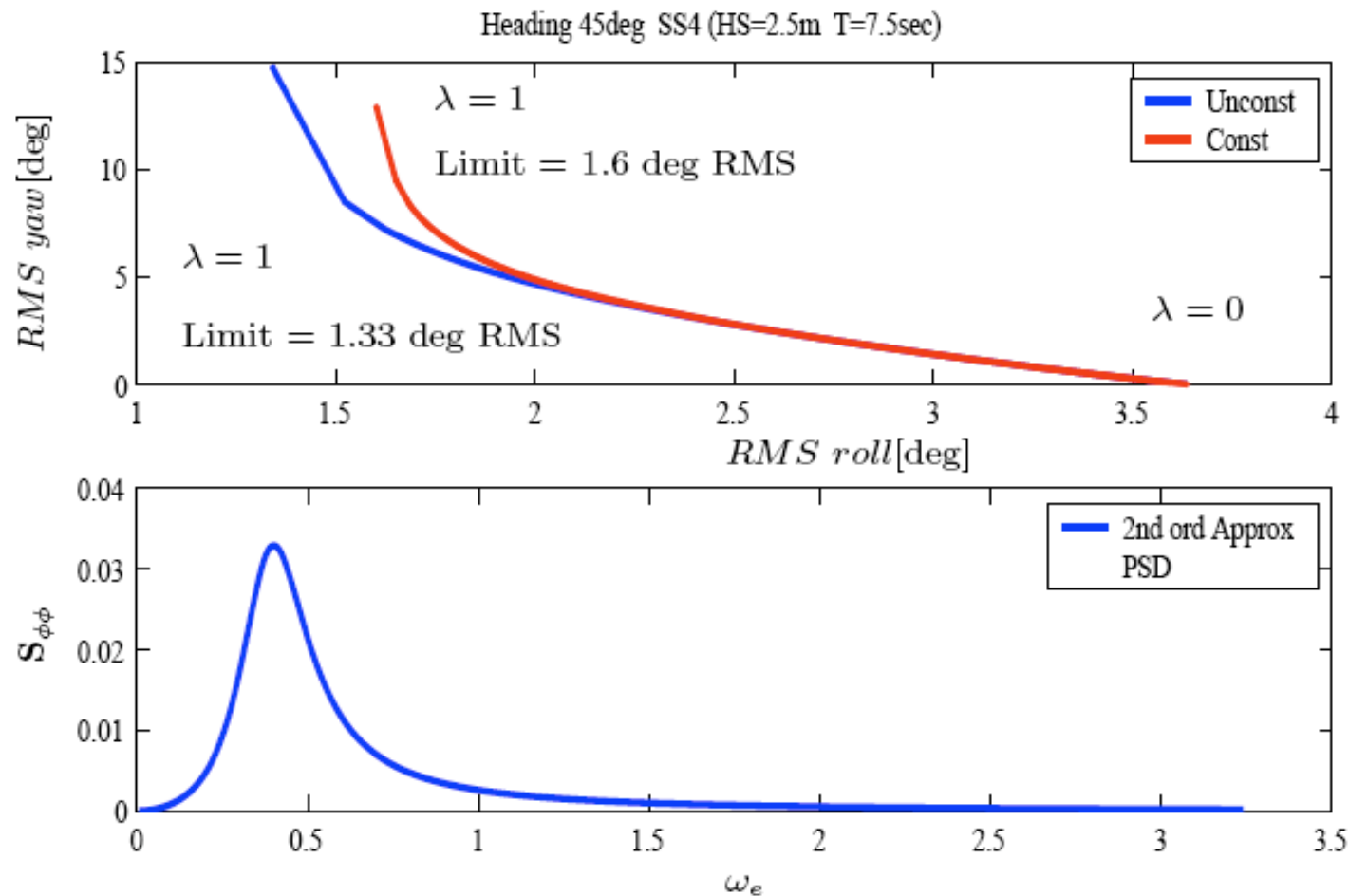
Performance Limitations – Perez et al. (2003)

IVC OCP: Rudder angle RMS limited to 15deg



Performance Limitations – Perez et al (2003)

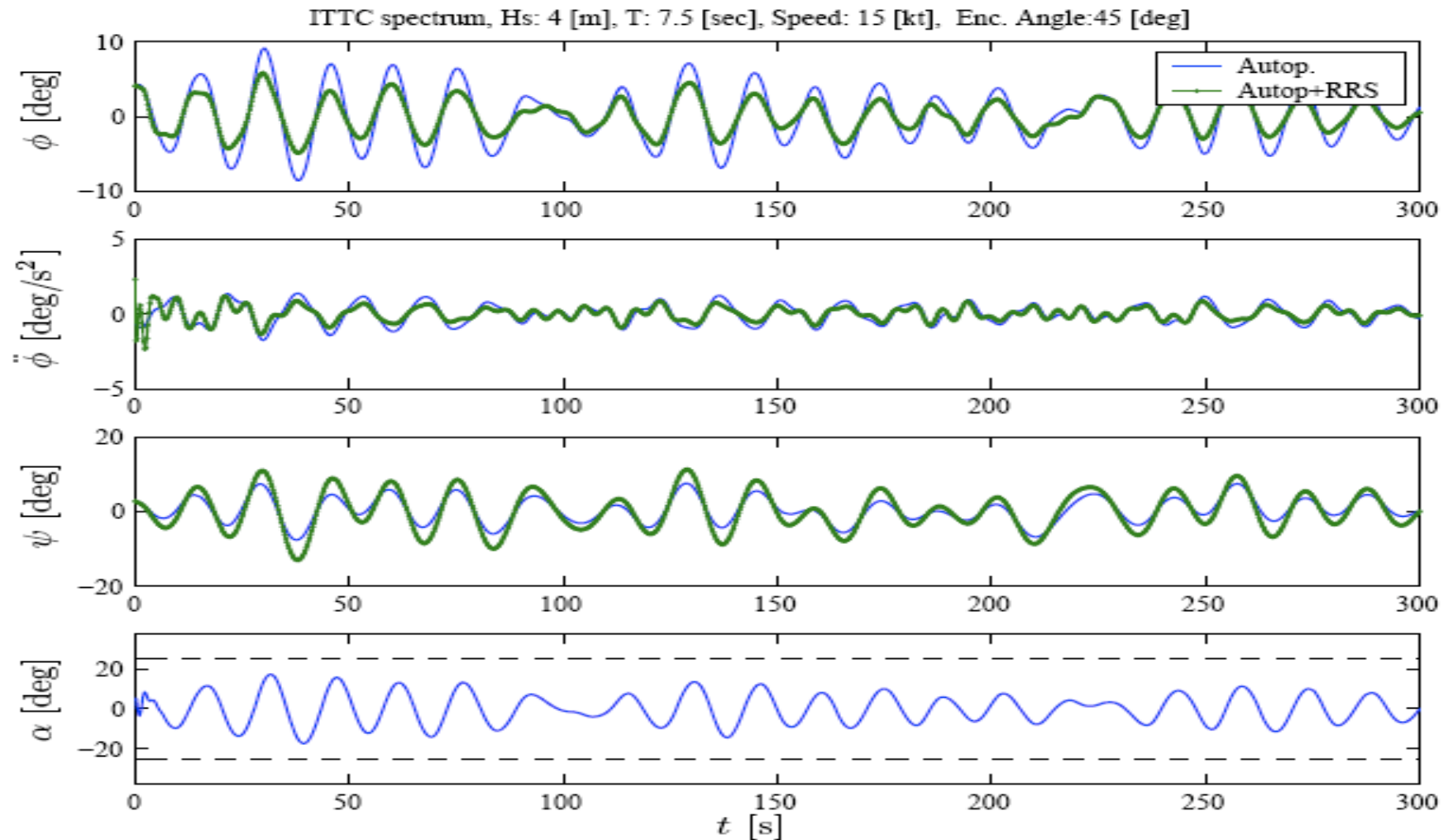
IVC OCP: Rudder angle RMS limited to 15deg



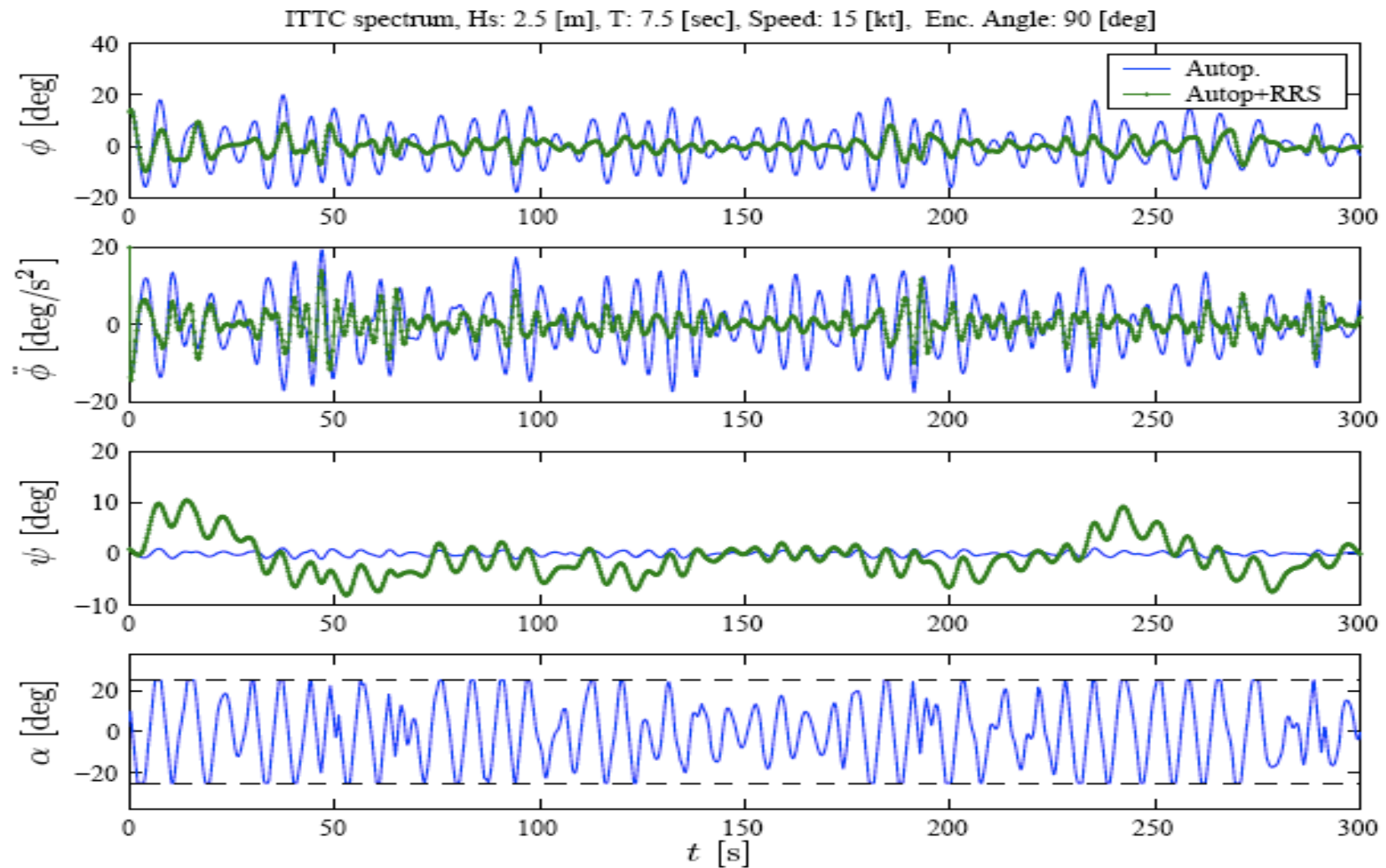
Performance Limitations

- ▶ At low encounter frequencies,
 - ▶ RR vs yaw interference can be large
 - ▶ MNP dynamics limits RR achievable performance.
- ▶ At high encounter frequencies,
 - ▶ Limitations of rudder machinery usually dominate RR achievable performance.

Performance Limitations (quatering seas RR 40%)



Performance Limitations (beam seas RR 64%)



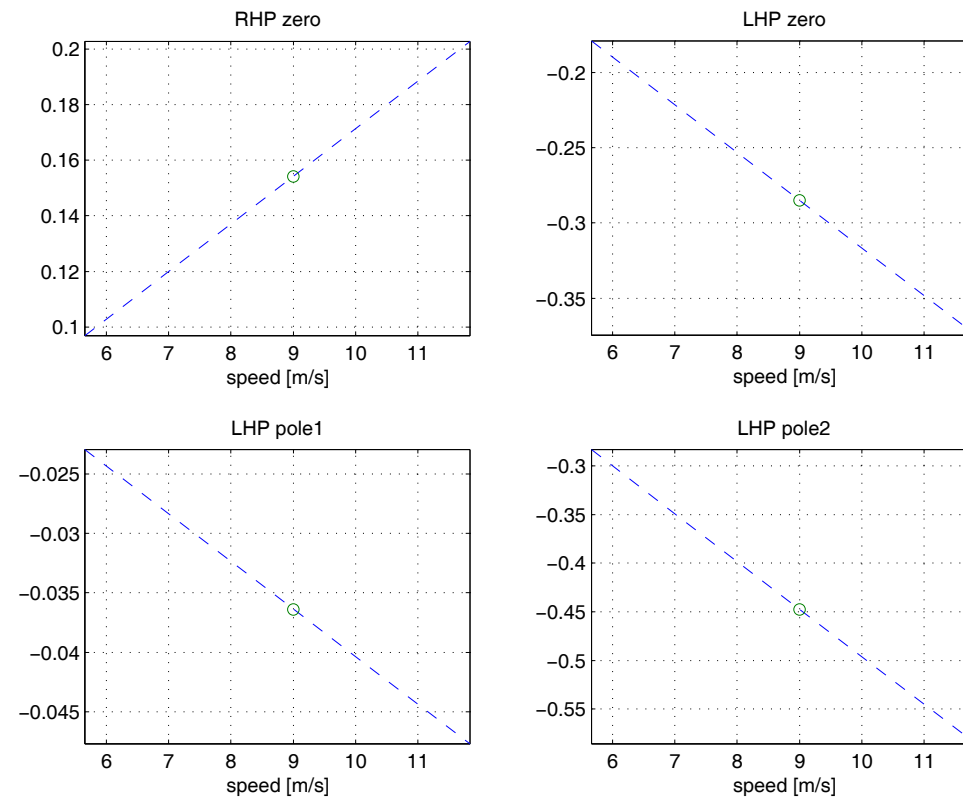
Control Strategies

- ▶ PID,
- ▶ Hinf
- ▶ Loop shaping
- ▶ Adaptive LQG
- ▶ Model Predictive Control
- ▶ Switched control
- ▶ Nonlinear control

Changes with Speed (Blanke & Christiansen 1993)

► Rudder to Roll TF:
$$G_{\phi\delta}(s) = \frac{c_{\phi\delta}(1 + s\tau_{z1})(1 - \frac{s}{q})}{(1 + s\tau_{p1})(1 + s\tau_{p2})(\frac{s^2}{\omega_p^2} + 2\zeta_p \frac{s}{\omega_p} + 1)}$$

Changes in dynamic response due to speed coupled with changes in disturbance spectrum results in a strong need for adaptation.



H2 – Optimal Design

$$\Phi_{\phi\phi}(\omega) = H_d^*(j\omega)H_d(j\omega)$$

$$E(\phi^2) = \|H_d - QV\|_2^2 \quad V = H_d G_{\delta\phi}$$

$$C_{\delta\phi}(s) = (G_{\delta\phi}^-)^{-1} \frac{H_d^{-1} H_d^-}{1 - \frac{s-q}{s+q} H_d^{-1} H_d^-}$$

$$G_{\phi\delta}^-(s) = G_{\phi\delta}(s) \frac{s+q}{s-q} \quad H_d(s) \frac{s+q}{s-q} = H_d^-(s) + H_d^+(s)$$

This shows the strong dependency on sea state, sailing conditions and changes in the vessel model --- adaptation is required.

Direct Sensitivity Specification (Blanke et al. 2000)

Targeted (Desired) Sensitivity:

$$S_d(s) = \frac{\frac{s^2}{\omega_d^2} + 2\zeta_d \frac{s}{\omega_d} + 1}{(1 + \frac{s}{\beta\omega_d})(1 + \frac{\beta s}{\omega_d})} \quad S(s) = (1 + C_{\delta\phi}(s)G_{\phi\delta}(s))^{-1}$$

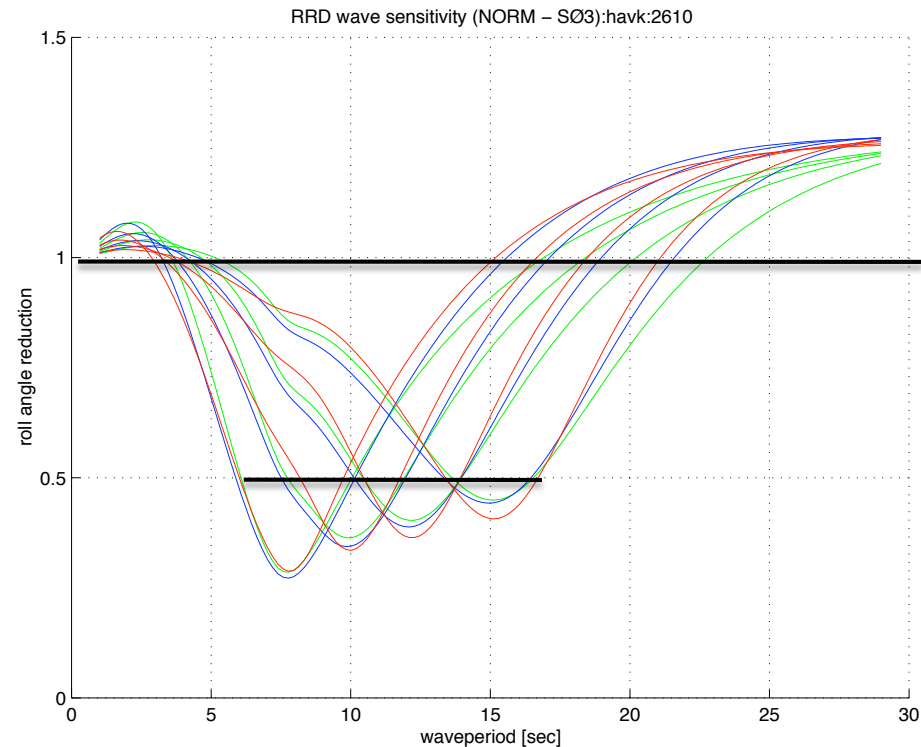


$$C_{\delta\phi}^s(s) = (S_d(s)^{-1} - 1)G_{\phi\delta}^{-1} \frac{1 - \frac{s}{q}}{1 + \frac{s}{q}} P(s)^{-1}$$

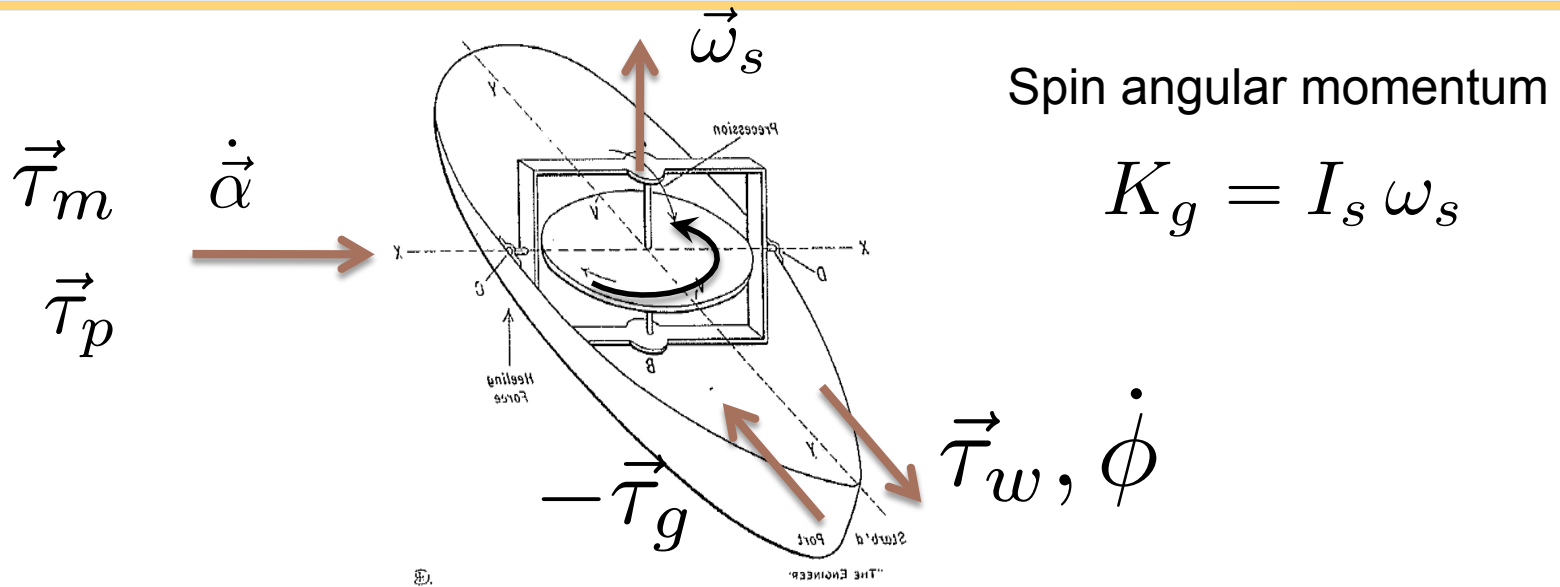
Direct Sensitivity Specification (Blanke et al. 2000)

$$C_{\delta\phi}^s(s) = \frac{k_1 s \frac{s^2}{\omega_p^2} + 2\zeta_p \frac{s}{\omega_p} + 1}{c_{\phi\delta\omega_d} \frac{s^2}{\omega_d^2} + 2\zeta_d \frac{s}{\omega_d} + 1} \frac{(1 + s\tau_{p1})(1 + s\tau_{p2})}{(1 + s\tau_{z1})(1 + \frac{s}{q})}$$

Successful performance was demonstrated for the SF300 patrol boat of the Danish Navy



Gyro Stabilisation



$$I_{44}\ddot{\phi} + B_{44}^l\dot{\phi} + B_{44}^n|\dot{\phi}|\dot{\phi} + C_{44}\phi = \tau_w - \underbrace{K_g\dot{\alpha}\cos\alpha}_{\tau_g},$$

$$I_g\ddot{\alpha} + B_g\dot{\alpha} + C_g\sin\alpha = \underbrace{K_g\dot{\phi}\cos\alpha}_{\tau_m} + \tau_p$$

Gyro as a Damper

$$\tau_p : \dot{\alpha} \approx q \dot{\phi}$$



$$I_{44}\ddot{\phi} + (B_{44}^e + \underbrace{nK_g q})\dot{\phi} + C_{44}\phi = \tau_w$$

Increase of roll damping
due to the gyrostabiliser

n - number of gyros
 q -constant

Gyro-control Design

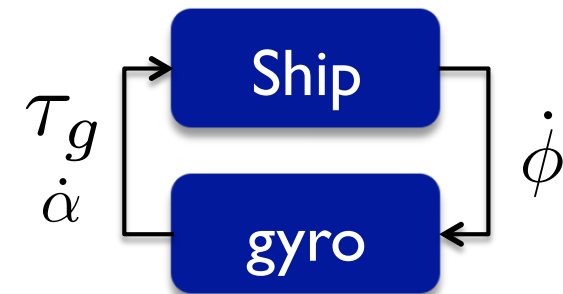
Gyro full state control: $\tau_p = -K_a \alpha - K_r \dot{\alpha}$

Roll to precession Transfer function:

$$G(s) = \frac{\alpha(s)}{\phi(s)} = \frac{\dot{\alpha}(s)}{\dot{\phi}(s)} = \frac{K_g s}{I_g s^2 + B'_g s + C'_g}, \quad \begin{aligned} B'_g &= B_g + K_r, \\ C'_g &= C_g + K_a \end{aligned}$$

The control design objective becomes:

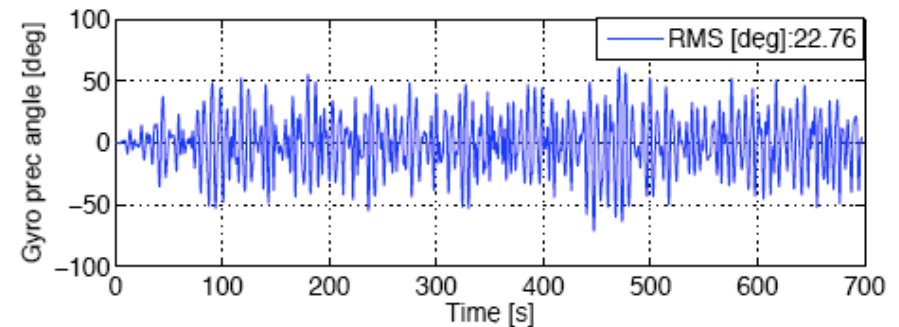
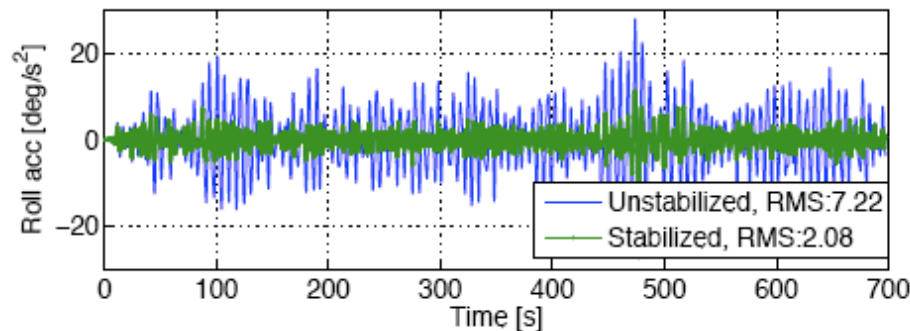
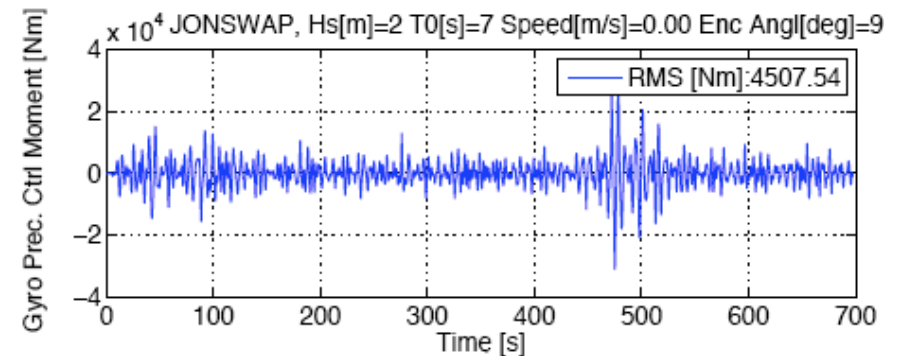
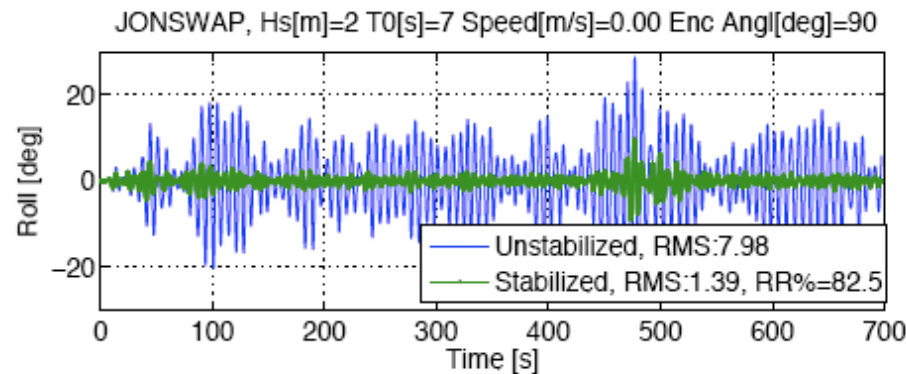
$$\tau_p : \dot{\alpha} \approx q \dot{\phi} \rightarrow \begin{aligned} |G(j\omega)| &\approx q, \\ \forall \omega \in \Omega \quad \arg G(j\omega) &\approx 0 \end{aligned}$$



This can be achieved by forcing two real poles on $G(s)$ (Perez Steinman, 2009)

Irregular Sea Performance (Perez & Steinamn 2009)

Roll Reduction in RMS: 82%
Vessel displacement: 360 ton
Gyro unit weight: 13 ton



IV - Research Outlook

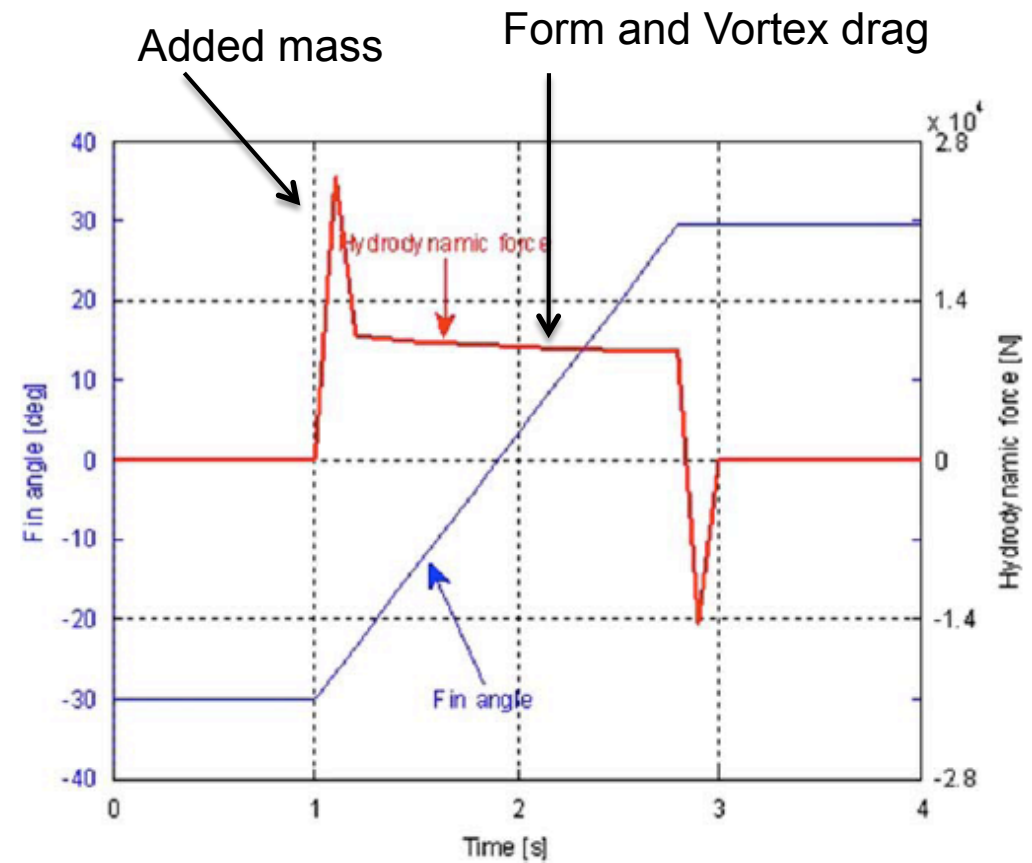
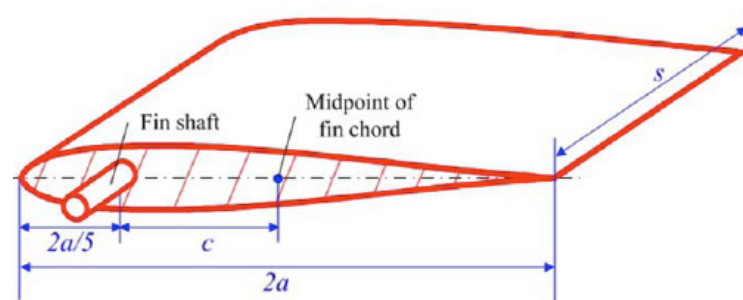


New Devices
Unsolved Problems

8th IFAC CAMS, Sept 15th-17th, Rostock,
Germany

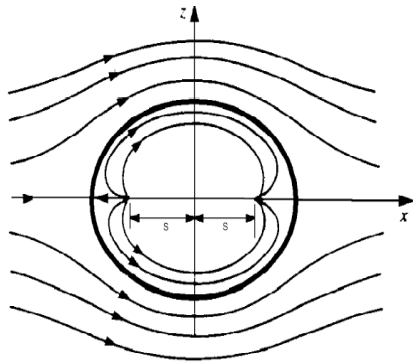
New Devices

Flapping fins (Fang et al. 2009 Ocean Engineering 36)

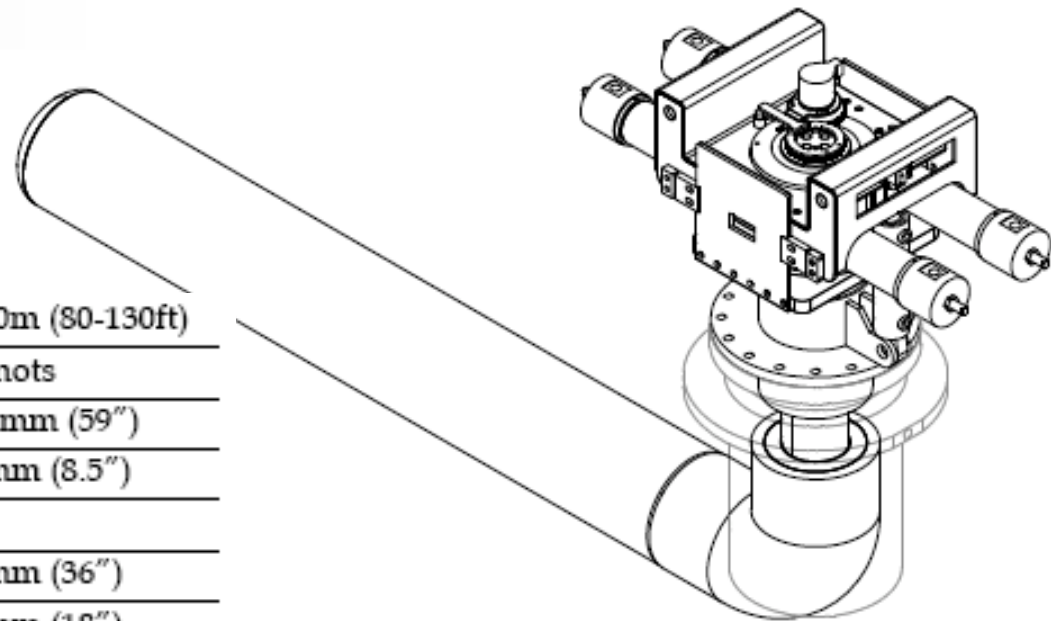


New Devices

Rotor Stabiliser (Quantumhydraulic.com)



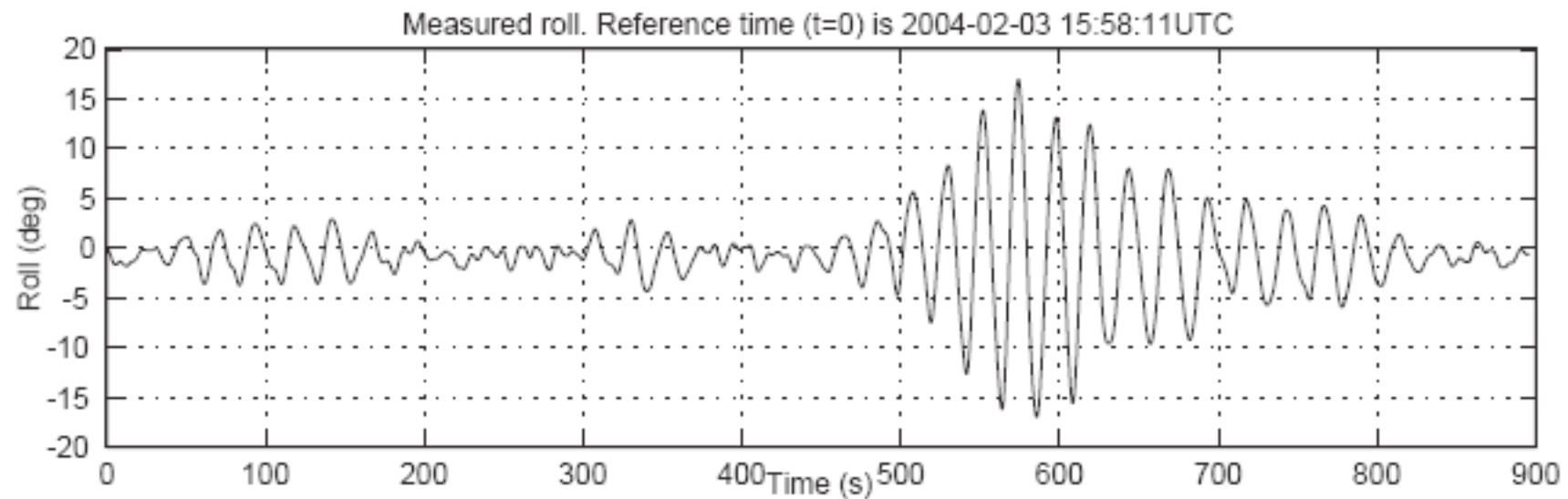
Typical Vessel Length*	25-40m (80-130ft)
Maximum Underway Operating Speed	14 knots
Rotor Length	1500mm (59")
Rotor Diameter	217mm (8.5")
Angular Travel (total mechanical)	150°
Length (inside vessel after installation)**	920mm (36")
Width**	450mm (18")
Height (overall)**	1110mm (44")
Height (inside vessel, retracted)**	890mm (35")



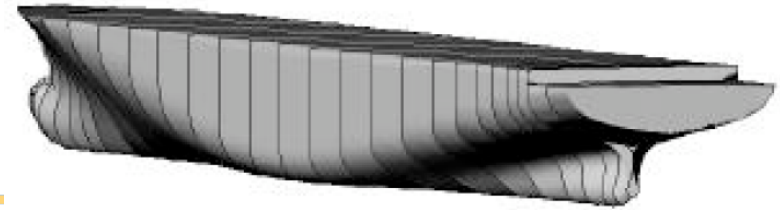
Parametric roll



Parametric roll is an auto-parametric resonance phenomenon whose onset causes a sudden rise in roll oscillations.



Conditions for PR



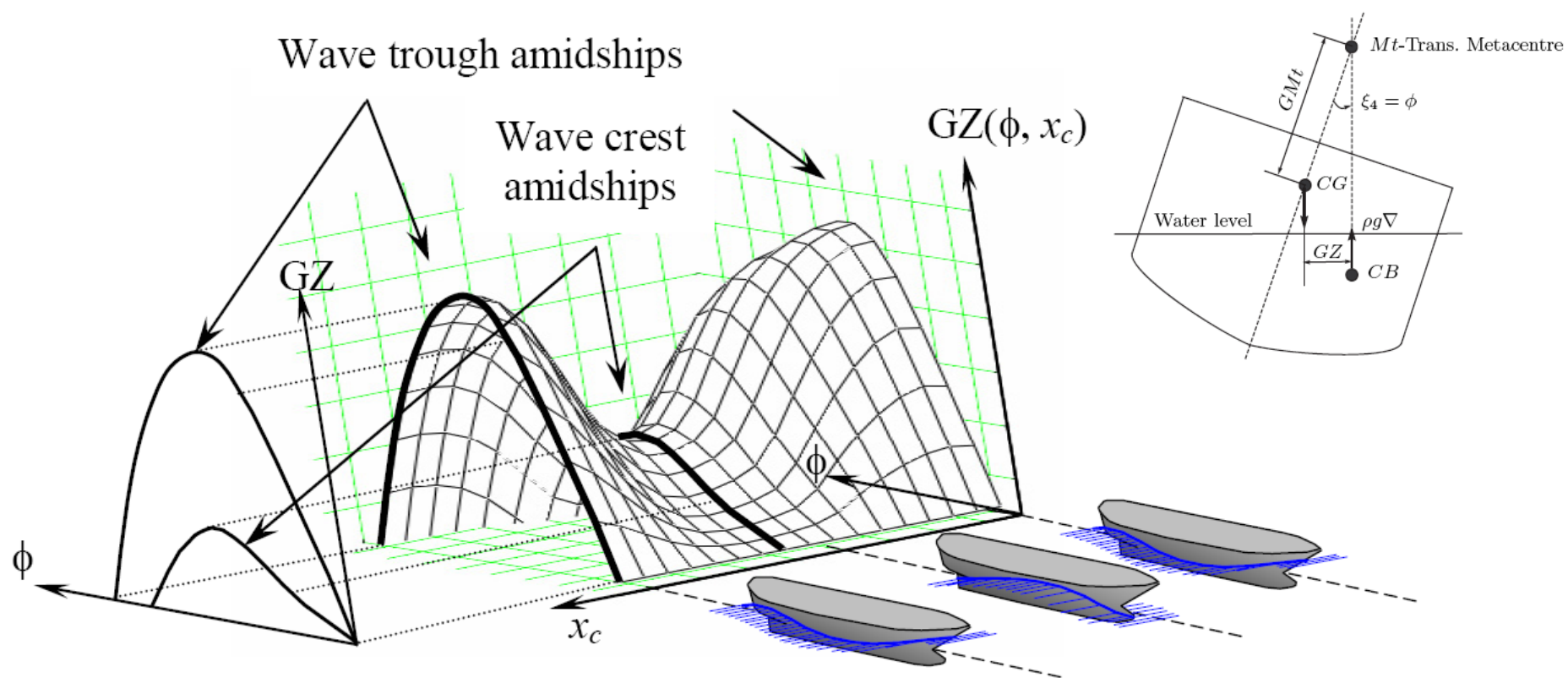
The following conditions can trigger the effect:

- Hull designs with significant bow flare and hanged stern
- Sailing in longitudinal waves
- Wave length close to the length of the vessel
- Encounter frequency is twice the roll natural frequency
- Low roll damping

Parametric roll is particular of modern container ships, cruise ships, car ferries, and fishing vessels.

Restoring in Waves – Dynamic Stability

$$K(\phi) = \rho g \nabla GZ(\phi)$$



Videos



Conclusion

- ▶ For over 100 years different devices have been proposed to control roll motion (reduction)
- ▶ Most of these devices require control systems to work
- ▶ Control design is not trivial due to
 - ▶ Fundamental limitations
 - ▶ Widely-varying disturbance characteristics
 - ▶ Actuator limited authority

Current State and Research Outlook

- ▶ New developments are being put forward by yacht industry
 - ▶ Gyros, Flap fins, rotor stabilisers (need for zero speed performance)
- ▶ Navies are still considering RRD
- ▶ Submarines need roll control at low speeds when in the surface

Research outlook

- ▶ Adaptation to sea state and sailing conditions still remains an issue
- ▶ New devices with interesting hydrodynamics
- ▶ Integrated vessel and roll control design (performance prediction)
- ▶ Parametric roll

Thank You



Death Roll of Sailing Vessels